

Inequality, Informality, and Optimal Progressivity*

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How should governments tax labor income when workers can shift into untaxed informal work? We combine Latin American household surveys with a heterogeneous-agent model featuring search frictions, savings, and sectoral labor supply. Negative income taxes draw low earners into formal work, while higher marginal rates at the top encourage informality. In our Mexican calibration, welfare-maximizing progressivity is nearly an order of magnitude above the current level. The reform raises steady-state welfare by 4.3% of consumption and reduces informality by 5.4 percentage points. Higher labor-tax levels also redistribute but increase informality; capital taxes redistribute with little effect on informality. At equal revenue, progressivity yields three to four times the welfare gain of either level tax. Higher earnings inequality substantially increases the welfare gains from progressivity while leaving its optimal level nearly unchanged.

JEL Codes: E26, H24, H26, O17, D31.

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1 Introduction

Should developing economies subsidize formal work through negative income taxes? While many OECD tax systems embed in-work subsidies at the bottom of the earnings distribution, such policies are rare in high-informality economies, where work-contingent transfers are often viewed as infeasible because much income is earned outside the tax net. Informality, however, is also an active margin of response to fiscal incentives. When households can allocate labor between taxed formal work and untaxed informal work, progressive taxation can draw low earners into the formal sector. The effects of these subsidies depend on access to formal jobs and on how other workers respond to the taxes needed to finance them. A reform therefore changes both the distribution of disposable income and the formal tax base. The central question is how far to tilt the tax schedule toward work-contingent subsidies at the bottom, given the offsetting evasion response at the top, and whether other tax instruments can achieve similar gains.

We study this question by combining household survey evidence with a static and a quantitative model. Our empirical analysis uses nationally representative surveys for Brazil, Colombia, Mexico, and Peru in 2023–24. The data reveal steep gradients in employment, unemployment, and informality across the income distribution. Informality is nearly universal at the bottom and remains common even in the highest income decile. Lower-income workers exit formal jobs more often and re-enter them more slowly. In Mexico, the rotating panel also allows us to compare the same workers’ earnings across sectors. For workers observed in both sectors, we estimate an elasticity of informal earnings with respect to their own formal earnings of approximately 0.8. Informal earnings thus rise less than one-for-one with formal earning power, so the earnings cost of moving into informality increases with ability. Together, these findings discipline the differences in access to formal employment and in returns to ability that shape households’ sectoral choices.

We first use a simple model to explain how progressive taxation affects those choices. Households allocate labor between formal employment, which is subject to unemployment risk, and informal work, which is always available but offers lower returns to ability. Informal work provides income when formal earnings are lost, while its lower returns make it increasingly costly for high-ability households to forgo formal work. Progressivity operates through two opposing forces. At the bottom, work-contingent subsidies increase the return to formal work and draw low earners into the tax base—an *inclusion margin*. At the top, higher marginal rates shift labor supply into the untaxed sector—an *evasion margin*. The static model generates hump-shaped welfare as progressivity in-

creases. We then embed the same labor-supply margin in a quantitative heterogeneous-agent model with search frictions and self-insurance through savings. Formal jobs must be found and can be lost, while productivity risk affects earnings in both sectors. Households adjust savings as well as labor supply, and wages, vacancies, and asset prices respond in general equilibrium. The model is calibrated to Mexican inequality, labor-market outcomes, and public finances.

Our first result is that Mexico can raise progressivity substantially while improving welfare and reducing informality. In our calibration, the welfare-maximizing progressivity parameter is 0.10, against 0.015 in the benchmark, nearly an order of magnitude higher. This parameter measures the fraction of differences in log formal earnings removed by taxation. Welfare increases by 4.3% of consumption, the informal share of hours worked falls by 5.4 percentage points, consumption inequality falls by 3.2%, and tax revenues rise by about half a percentage point of GDP. These are steady-state welfare gains that account for both consumption and leisure. The reform holds government spending at its calibrated share of output and returns changes in revenue net of spending to households through the lump-sum fiscal instrument. The implied tax schedule subsidizes low formal earnings while raising marginal rates at the top. The marginal tax rate becomes negative in the bottom decile of formal earnings and reaches 29% in the top decile, compared with a nearly flat baseline schedule. The welfare-maximizing degree of progressivity coincides with the level that maximizes revenue as a share of GDP. Up to this level, redistribution draws labor into formal work and expands the tax base. Beyond the optimum, evasion erodes the tax base, and welfare and revenues decline together.

Our second result is that progressivity combines redistribution and formalization in a way that increases in other income taxes do not. We vary one instrument at a time under the same government budget rule, holding the other tax-rate parameters fixed. Raising the labor-tax level can also improve welfare when its proceeds are returned to households. At its welfare-maximizing level, ten percentage points above the benchmark, it delivers a gain of 6.3% of consumption, but informality rises by almost 20 percentage points. Higher capital income taxation generates welfare gains of up to 1.6% over the range we consider, with little change in informality and welfare still rising at the upper edge of the grid. Progressivity is the only instrument that raises welfare while substantially reducing informality, and it delivers the largest gain for a given amount of revenue: a higher labor-tax level that raises half a percentage point of GDP yields 1.4% of consumption, a higher capital tax 1.2%, against 4.3% for progressivity. The distinction is the way each instrument redistributes. A higher labor-tax level finances redistribution while reducing the return to formal work throughout the

earnings distribution. Capital taxation redistributes with little effect on sectoral labor allocation. Progressivity instead raises the return to formal work at the bottom and finances the subsidy through higher taxation at the top. Its welfare gains also survive when additional revenue is absorbed by government consumption: in that case, raising labor- or capital-tax levels no longer improves welfare.

Our third result concerns the relationship between inequality and optimal progressivity. Higher idiosyncratic productivity risk strengthens both margins. Low earners have more to gain from subsidies to formal work, while taxable income becomes more concentrated among high earners who can shift labor into informality. A more dispersed earnings distribution therefore increases both the scope for redistribution and the importance of the evasion response. In our quantitative experiments, these forces leave the welfare-maximizing progressivity nearly unchanged, while the gain from adopting it rises from 2.8% to 6.6% of consumption between the low- and high-inequality economies. This differs from the usual prescription that greater uninsurable earnings risk calls for more progressivity (Heathcote et al., 2017, 2020). Here the stronger insurance motive is met by an evasion response that also strengthens with inequality, so the optimal degree of progressivity depends on how the two interact.

Fixing government spending in levels gives the same optimal progressivity as the benchmark, with a welfare gain of 3.0% of consumption. We also use the Mexican panel estimates to assess the role of earnings risk and informal returns. The estimated transitory component of earnings variance is 2.4 times larger in informal work. This distinction matters because households can insure transitory fluctuations more readily through savings than persistent earnings differences. Recalibrating the model with a common transitory shock and sector-specific loadings, or with the estimated informal returns, leaves optimal progressivity several times above its benchmark. These alternatives change the magnitude of the gains and the fit of the wealth and earnings distributions. They support substantial scope for increasing progressivity under the maintained risk assumptions, while showing that the size of the welfare gain depends on earnings dynamics as well as the use of tax revenue.

Related Literature. Our paper connects optimal redistribution with incomplete markets to the macro-development literature on informality. Quantitative heterogeneous-agent models show how the tax code provides insurance and redistributes across households (Conesa and Krueger, 2006; Heathcote et al., 2017; Kindermann and Krueger, 2022), and how progressivity affects tax revenues (Holter et al., 2019). We adopt the parametric tax framework of Heathcote et al. (2014, 2017).

Closest in this literature, [Ferriere et al. \(2023\)](#) show that implementable progressive tax systems can approach second-best allocations in fully formal economies. We make the tax base endogenous through formal–informal labor reallocation and compare progressivity with changes in labor- and capital-tax levels.

[Doligalski and Rojas \(2023\)](#) study optimal redistribution when workers can simultaneously supply formal and informal labor. Their Mirrleesian analysis shows how informal responses constrain redistribution. We study an implementable progressive schedule in a dynamic general-equilibrium environment with search frictions and precautionary saving. Our focus is on the range over which progressivity jointly improves welfare and formalization, its performance relative to other fiscal instruments, and how that range changes with inequality. On the consumption side, [Bachas et al. \(2024\)](#) show how informal purchases shape tax incidence and redistribution. We study labor informality as an endogenous response to the personal income tax schedule.

Our labor-market framework builds on dual-sector search models studying minimum wages and enforcement ([Albrecht et al., 2009](#); [Bosch and Esteban-Pretel, 2012](#)), on-the-job search ([Bobbia et al., 2021](#)), and wage sorting ([Meghir et al., 2015](#)). A complementary literature emphasizes firms’ decisions to register and hire off the books ([Ulyssea, 2018](#)), compliance costs ([Kleven et al., 2016](#)), and firm organization over development ([Engbom et al., 2025](#)). Firm decisions and household labor choices jointly shape informality. We study how households divide labor between the formal jobs available to them and informal work. Workers move between sectors throughout the income distribution, and households often combine both. Personal income taxation directly changes their returns to formal work. Holding registration costs, corporate taxation, and enforcement fixed, we quantify how this household response shapes redistribution, formalization, and the relative performance of tax instruments.

2 Empirical Evidence: Informality Across Incomes

We use household surveys from Brazil, Colombia, Mexico, and Peru to document how labor-market outcomes vary across the income distribution. Workers in lower-income households are more likely to work informally and face weaker attachment to formal employment. We then use the Mexican panel to estimate returns to informal work, to separate persistent from transitory earnings dispersion in each sector, and to document how workers and households combine formal and informal employment. These facts inform the structure and calibration of the quantitative model in [Section 4](#).

2.1 The Data

We use nationally representative household surveys from Brazil (PNADC, 2024), Colombia (GEIH, 2023), Mexico (ENOE, 2024), and Peru (ENAH, 2023). All four record labor earnings, labor-force status, informality identifiers, and demographics. Employment and unemployment follow each survey’s standard definitions; informality follows country-specific criteria based on registration, social-security coverage, and establishment characteristics. Appendix A describes these definitions, and Appendix F reports additional survey details and earnings and education patterns.

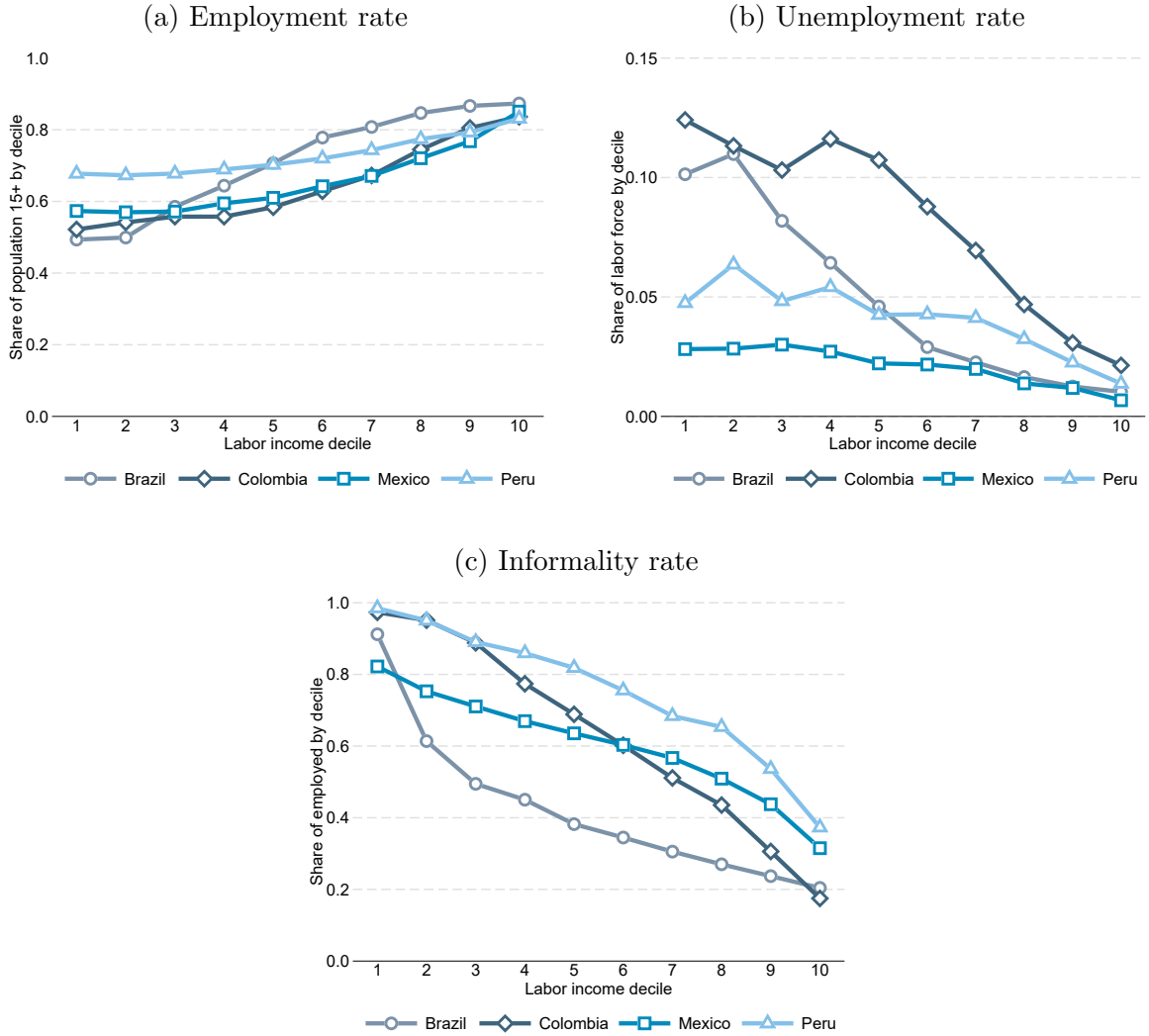
Our cross-country analysis covers individuals aged 15 and older in households with positive labor earnings. We construct country-specific deciles of household per-capita labor earnings, defined as the sum of household members’ reported monthly labor earnings divided by household size. This measure excludes capital income and transfers. All estimates use survey weights. Labor earnings are highly unequal: the Gini ranges from 0.42 in Mexico to 0.58 in Colombia (Table F.1).

2.2 Labor Market Outcomes Across the Income Distribution

Figure 1 shows employment, unemployment, and informality by labor-income decile. Individuals at the bottom are less likely to be employed and more likely to work informally than those at the top. Employment rates in the bottom decile are around 50% in Brazil, Colombia, and Mexico, and rise substantially toward the top decile. Unemployment generally declines with income, with larger differences between the bottom and top in Brazil and Colombia. Informality is high at the bottom and remains common at the top.

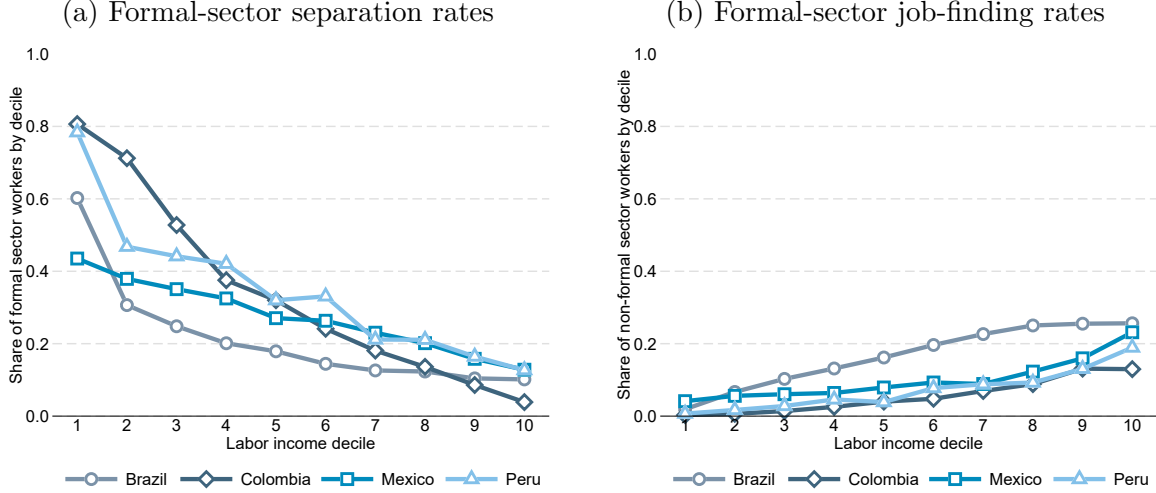
We use the panel structure of the Brazilian, Mexican, and Peruvian surveys, and retrospective questions in Colombia, to compute transitions into and out of formal jobs. Figure 2 shows that workers in lower-income households are less likely to find formal jobs and more likely to lose them. These gradients describe unequal attachment to formal employment and guide the productivity-specific transition rates in the quantitative model.

The transition gradients describe differences in access to formal jobs and in the duration of formal attachment. At the same time, informal work in the top income decile shows that informality extends beyond households with the weakest labor-market outcomes. These patterns motivate a model combining formal-sector search frictions with households’ choices between formal and informal work.



Notes: The sample covers individuals aged 15 and older in households with positive labor earnings. Deciles are constructed separately by country using household per-capita labor earnings. Survey years are 2024 for Brazil and Mexico and 2023 for Colombia and Peru. All estimates use survey weights. Informality follows country-specific definitions based on registration, social-security coverage, and establishment characteristics; see Appendix A.

Figure 1: Labor market indicators by labor income decile, 2023/24



Notes: Annual transition probabilities for individuals aged 15 and older in households with positive labor earnings. Separation means leaving formal employment for informal employment, unemployment, or nonparticipation; job finding means entering formal employment from these states. Deciles are constructed separately by country using household per-capita labor earnings. Survey years are 2024 for Brazil and Mexico and 2023 for Colombia and Peru. All estimates use survey weights. Country-specific formality definitions and survey details are given in Appendix A.

Figure 2: Annual formal-sector separation and job-finding rates, 2023/24

2.3 A Closer Look at Mexico: the ENOE Rotating Panel

The ENOE interviews each household for five consecutive quarters, allowing us to follow workers across jobs, sectors, and earnings realizations. We link the 2021–2024 panel, using a window after the survey’s redesign. The Mexican cross-sectional facts above use 2024; the earnings-variance and sector-mixing estimates below use the linked panel. The linked file contains about 5.1 million person-quarter observations on 1.6 million individuals. The wage estimates use hourly earnings from the main job.

Informal returns to ability. The sectoral technology in Sections 3 and 4 implies that a worker with ability h earns $W^f = w^f h$ per hour in formal work and $W^s = w^s \Xi h^\xi$ in informal work. For a given ability, the two log wages are related by

$$\ln W_i^s = a + \xi \ln W_i^f, \quad a = \ln \Xi + \ln(w^s/(w^f)^\xi). \quad (1)$$

The slope ξ measures how informal earnings change with formal earning power. A value below one means that the earnings advantage of formal work increases with ability. The intercept combines informal productivity with equilibrium wages, so it does not identify the scale parameter Ξ separately.

We estimate this relationship among workers observed in both sectors, instrumenting their formal log wage with a second formal observation of the same worker at a different date. The sample requires at least one informal and two formal observations within the five interviews. Comparing earnings of the same worker avoids interpreting cross-sector differences in worker composition as differences in returns to ability. Within-person ordinary least squares yields an elasticity of 0.36; the instrumental-variables estimates address attenuation from serially uncorrelated measurement error.

The instrument must be uncorrelated with intervening ability innovations among the workers who switch sectors. Selection on these innovations can therefore affect the interpretation of the estimated slope. Mean reversion also matters when the two wage observations are at different distances from the instrument. The baseline estimate adjusts for these differences using estimated quarterly persistence; before removing observable characteristics, the annual persistence of log wages is 0.944. We also report a lag-matched sample in which the formal and informal observations are equally distant from the instrument and no adjustment is required. Appendix A.3 describes the sample, controls, and estimator.

Table 1 reports the main estimates. The elasticity, which corresponds to the curvature of informal returns to ability ξ in Sections 3 and 4, is approximately 0.8 and precisely estimated below one. Among these switchers, informal earnings rise less than one-for-one with formal earning power. The lag-matched estimate is close to the baseline estimate, while including informal self-employment gives a somewhat lower elasticity. For salaried workers and lag-matched observations, the joint specification implies a wage ratio at mean formal earnings of 0.92. Appendix Table A.1 also reports alternative wage concepts and informality definitions; the elasticity estimates range from 0.75 to 0.87. The baseline model calibrates ξ to sectoral earnings moments; Appendix C fixes it at the panel estimate, with persistence 0.944, in a robustness calibration.

A further pattern in the panel is consistent with lower informal returns to ability. Relative to workers who start from the same formal-wage decile and remain formal, workers moving into informality lose 0.07–0.10 log points of earnings over a year in the second through sixth deciles, rising to 0.35 log points in the top decile. The earnings loss associated with moving into informality thus increases with initial formal earnings.

Persistent and transitory earnings variance. We split each sector’s cross-sectional variance of log earnings into a persistent and a transitory component. The persistent component is iden-

Table 1: Informal Returns to Formal Earning Power

Specification	Elasticity	Standard error
Salaried workers: baseline IV	0.840	0.027
Salaried workers: lag-matched IV	0.827	0.036
Including informal self-employment: joint IV	0.784	0.022

Notes: ENOE linked panel, 2021–2024, workers aged 25–60. Informal log hourly earnings are regressed on formal log hourly earnings, instrumented by a second formal observation of the same worker. The salaried sample contains 18,990 switchers and 102,102 wage triples, including 14,960 lag-matched triples. Lag matching places both wage observations equally far from the instrument. The first two rows residualize earnings within sector; the last adds informal self-employment (31,048 switchers), uses common covariate coefficients, and jointly estimates the slope and intercept. Only baseline IV adjusts for mean reversion. Estimates use survey weights and worker-clustered bootstrap standard errors. Appendix A.3 gives the construction and identifying assumptions.

tified from autocovariances at lags one and beyond, which are unaffected by serially uncorrelated measurement error. The estimated transitory component is 2.4 times larger in informal work: of the formal sector’s cross-sectional log-earnings variance of 0.325, 0.132 is transitory; of the informal sector’s 0.674, 0.313 is. Of the gap of 0.349, 0.169 is persistent and 0.180 transitory. Section 4 targets the persistent part of the gap, and Appendix C adds the sectoral transitory variances to the model in a robustness calibration.

Combining formal and informal work. In the ENOE panel, 2.4% of employed workers aged 25–60 hold jobs in both sectors, almost always a formal main job and an informal second job. This within-person allocation is the margin represented in our single-earner model. Across household members, sector mixing is more common: 22.8% of working households have members in both sectors, rising to 39.9% among households with at least two employed members. These household shares capture different members working in different sectors.¹

3 A Simple Model of Progressive Taxation with Informal Work

How can fiscal policy address the challenges of income inequality in economies with large informal sectors? To build intuition, we present a static model with heterogeneous households and an endogenous formal–informal labor-supply margin. Households allocate their labor between salaried formal

¹Individual mixing is measured by the sector of the main and second jobs; the second job’s sector is observed only in first-quarter waves. Household mixing uses all employed members aged 15 and older, classified by their main job, and includes households with at least one employed member. Both measures use survey weights from the 2021–2024 ENOE panel.

work and informal self-employment. Formal jobs offer higher returns to ability but expose households to unemployment risk, while informal work provides an always-available source of income. Informality captures subsistence entrepreneurship and tax avoidance (Schoar, 2009).

The government taxes only formal earnings. Progressivity affects formal labor supply through two channels that differ across households. At the bottom of the distribution, negative income taxes raise the return to formal work and draw low earners into the tax base—an *inclusion margin*. At the top, higher marginal rates discourage formal labor and make untaxed work more attractive—an *evasion margin*. The model shows how these opposing responses shape the welfare effects of progressive taxation.

Households. The economy consists of a continuum of households endowed with exogenous ability h , drawn from a log-normal distribution, $\ln h \sim \mathcal{N}(-\sigma_h^2/2, \sigma_h^2)$. Average ability is normalized to one. At the beginning of the period, households choose total labor supply $n \in [0, 1]$ and the fraction $\varpi \in [0, 1]$ allocated to formal work. The remaining fraction is allocated to informal self-employment. Formal labor earns $w^f h n \varpi$, but with probability q the household does not find a job and receives no formal earnings. This risk is common across ability levels and does not depend on ϖ : allocating more time to formality changes earnings if a job is found, but leaves the probability of finding one unchanged.²

Informal work faces no matching frictions and earns $w^s \Xi h^\xi n(1 - \varpi)$, where $\Xi < 1$ and $\xi < 1$. Earnings increase with ability in both sectors, but the informal technology compresses ability differences. Low-ability households therefore have a smaller opportunity cost of forgoing formal work, while high-ability households have more to gain from formal earnings. Informality in this model refers to self-employment, although informal salaried jobs are also widespread in developing economies.³

The government taxes formal earnings y through the schedule (Benabou, 2002; Ferriere et al., 2023; Heathcote et al., 2017)

$$T(y) = y - \tau^L y^{1-\tau^P}. \quad (2)$$

Here τ^P controls progressivity and τ^L scales after-tax formal earnings. Without progressivity, the proportional tax rate is $1 - \tau^L$; lowering τ^L raises the tax burden. With positive progressivity, the schedule allows negative income taxes at low earnings and positive taxes at higher earnings. Informal

²This probability differs from the vacancy-filling rate $q^v(\theta)$ in the quantitative model of Section 4.

³Labor informality extends beyond self-employment (Ulyssea, 2018). Much salaried informal employment is concentrated in small unregistered firms operated by self-employed individuals (La Porta and Shleifer, 2014).

earnings are untaxed, reflecting the difficulty of observing income from informal work (Banerjee et al., 2023). Households may also receive a lump-sum transfer τ^T regardless of employment status.

Negative income taxes are paid only when households earn formal income. They therefore tie redistribution to formal work: a household receives the subsidy by reallocating labor into the taxed sector. The lump-sum transfer, by contrast, enters the budget in both employment states and provides income even to households working entirely informally.

Given ability h , households choose consumption and labor supply to maximize expected utility

$$u(c^f, c^s) = (1 - q) \ln c^f + q \ln c^s - \frac{n^{1+\varphi}}{1 + \varphi}, \quad (3)$$

where φ governs the curvature of labor disutility. Consumption satisfies the budgets when households find and do not find a formal job,

$$c^f = w^s \Xi h^\xi n (1 - \varpi) + \tau^L (w^f h n \varpi)^{1-\tau^P} + \tau^T, \quad (4)$$

$$c^s = w^s \Xi h^\xi n (1 - \varpi) + \tau^T. \quad (5)$$

Labor is allocated before formal-job uncertainty is resolved. Informal earnings are available in both states, so informal work insures households against the loss of formal earnings. A household that works only informally has the same consumption in both states, but forgoes any gain from access to formal employment.

Production. A competitive firm combines aggregate formal and informal labor through the CES technology

$$Y = \left[(L^f)^{\frac{\epsilon-1}{\epsilon}} + (L^s)^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}}, \quad (6)$$

where ϵ is the elasticity of substitution. Wages equal marginal products. We abstract from aggregate productivity differences across sectors to focus on household skills. Reallocation between formal and informal work changes their relative scarcity and wages, feeding back into households' decisions. The quantitative model also allows sector-specific productivity.

Government. The government balances its budget, using formal-income taxes to finance spending G and transfers τ^T . In the experiments below, transfers are zero and τ^L adjusts so that spending remains 15% of output in every simulation. Changes in progressivity therefore alter the shape of the tax schedule while preserving the government's share of output. In the quantitative progressivity

experiment, the spending share and τ^L are held fixed, while τ^T adjusts to return additional revenue or collect any shortfall; Appendix D.2 examines alternative budget closures. The government budget and equilibrium conditions for the static model appear in Appendix E.

3.1 Informal Job Decisions

Households compare the return to formal work with its unemployment risk and the earnings forgone in informal work. At an interior allocation, their choice satisfies

$$(1 - q) \frac{(1 - \tau^P) \tau^L (w^f h n)^{1 - \tau^P} \varpi^{-\tau^P} - w^s \Xi h^\xi n}{c^f} = q \frac{w^s \Xi h^\xi n}{c^s}. \quad (7)$$

The left-hand side is the expected marginal utility gain from reallocating labor to formality when a job is found. The right-hand side is the expected loss of informal earnings when it is not. Because consumption is lower without a formal job, households value informal earnings particularly highly in that state. Their allocation depends on both relative earnings and the insurance informal work provides.

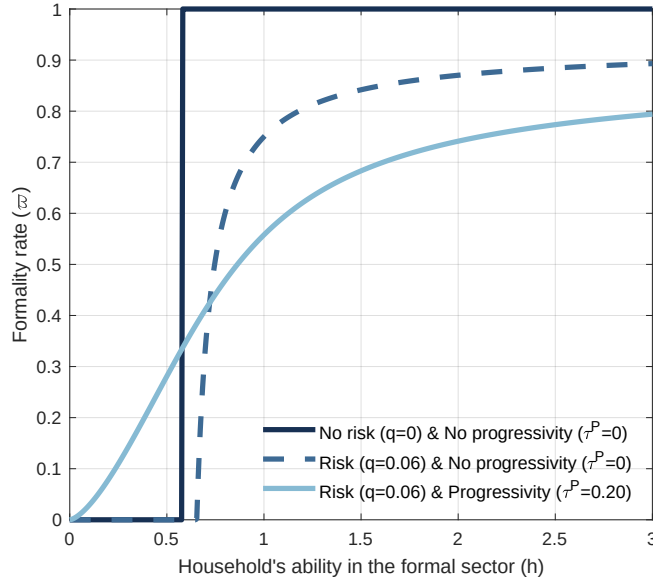
Without progressivity, low-productivity households choose only informal work when

$$h < \left(\frac{w^s \Xi}{(1 - q) \tau^L w^f} \right)^{\frac{1}{1 - \xi}}.$$

At given wages, higher unemployment risk or a higher proportional labor tax raises this cutoff. These households find formal work unattractive even when jobs may be available. Other households allocate labor to both sectors to self-insure against unemployment, accepting lower earnings when a formal job is found in exchange for income when it is not. The model thus captures both exclusion from formal work and voluntary informal employment (La Porta and Shleifer, 2014; Perry et al., 2007; Ulyssea, 2020).

Informality can arise even without taxes or unemployment risk. Households whose formal earnings per hour are below informal earnings, $w^f h < w^s \Xi h^\xi$, choose informal work. Taxes and unemployment risk expand this group by reducing the expected gain from formality. A subsidy to low formal earnings works in the opposite direction.

Figure 3 illustrates these choices across the ability distribution. Without unemployment risk or progressivity, households specialize in the sector with the higher return: low-ability households work informally and high-ability households work formally. Introducing unemployment risk changes the



Notes: Households choose formal labor share ϖ before job availability is known; q is the probability of no formal job and τ^P controls tax progressivity. Wages and the tax scale are held fixed, and lump-sum transfers are zero.

Figure 3: Formal-sector labor supply (ϖ)

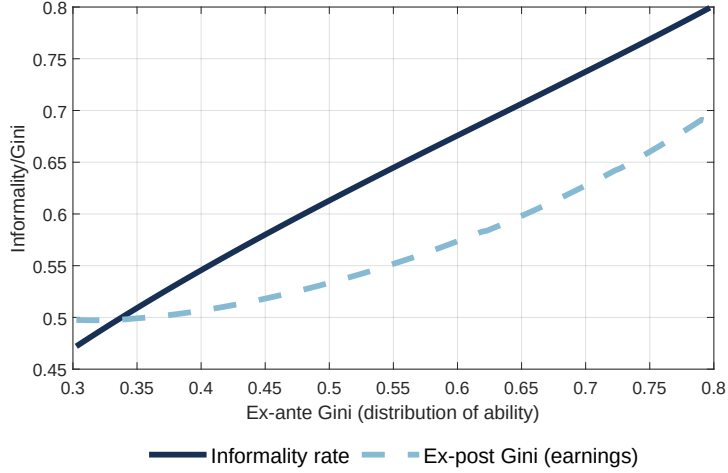
high-ability allocation. These households retain some informal work to insure against not finding a formal job, while low-ability households remain informal. In the illustrated economy, the formal share of labor still rises with ability, reflecting the increasing earnings advantage of formal work.

Progressivity changes the allocation in opposite directions at the bottom and the top. Negative income taxes encourage low-ability households to allocate part of their labor to formal jobs, strengthening the inclusion margin. High-ability households face higher marginal rates and shift labor toward untaxed work, strengthening the evasion margin. Informality therefore need not rise uniformly when taxation becomes more progressive. Its aggregate response depends on whether the increase in formal work among low earners outweighs the reduction among high earners.⁴

The figure holds wages fixed to isolate households' allocation decisions.⁵ In the policy experiments that follow, wages adjust with aggregate formal and informal labor. The response of the tax base and the resources available for consumption therefore incorporates these equilibrium changes in relative returns.

⁴We abstract from non-monetary job attributes, including flexibility and independence in self-employment (Maloney, 2004) and mandated formal-sector benefits (Levy, 2008). These could enter the allocation condition through additional preference terms (Galiani and Weinschelbaum, 2012; Gerard and Gonzaga, 2021).

⁵For this illustration, $w^f = 1.04$, $w^s = 1.26$, $\tau^L = 0.8$, and $\Xi = \xi = 0.5$.



Notes: Ability dispersion increases while mean ability remains one. Tax progressivity and lump-sum transfers are zero. The tax scale adjusts in each static-model equilibrium to finance government spending equal to 15% of output; other structural parameters are unchanged.

Figure 4: Inequality and informality

3.2 Inequality and Informality

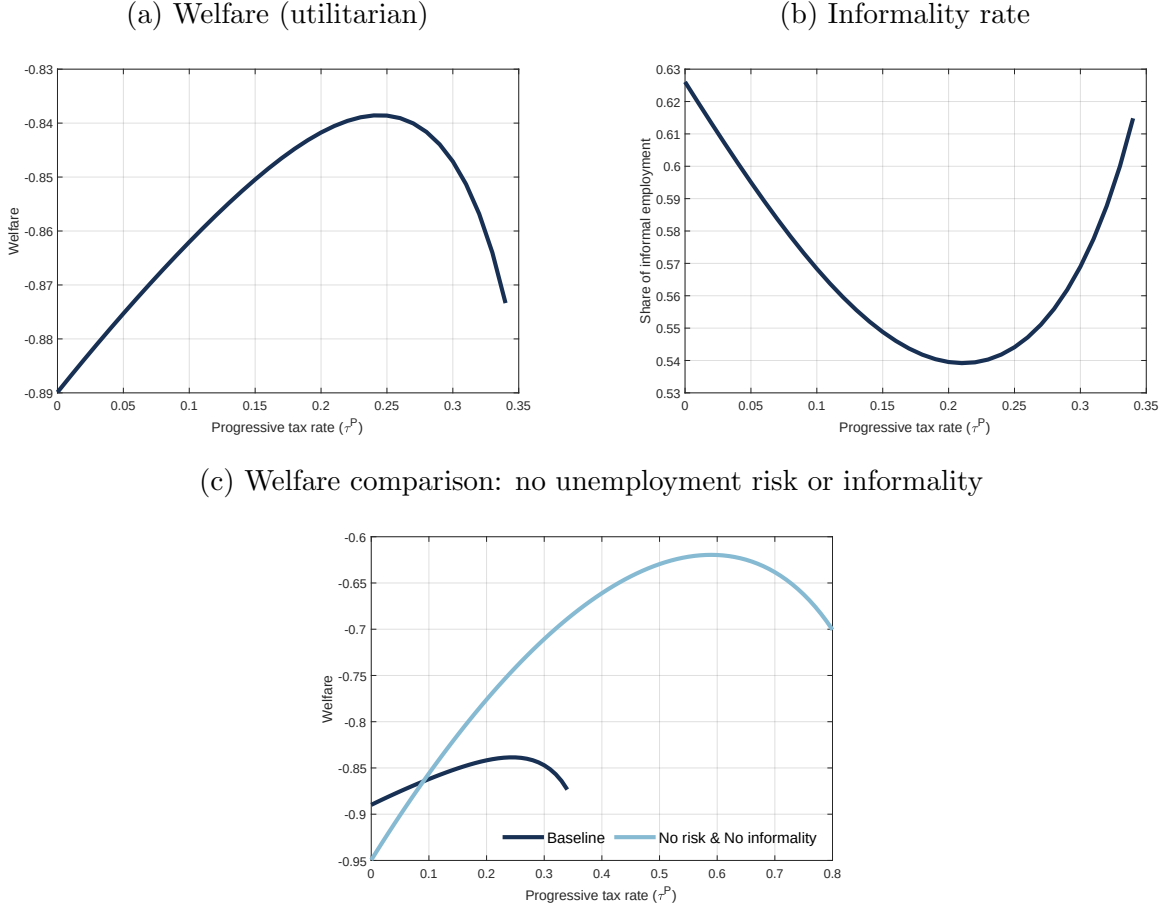
Households use informal work to insure against unemployment and avoid taxation. The insurance motive is stronger for low-ability households, whose lower formal productivity reduces the expected gain from formal employment. Changes in the ability distribution therefore affect informality and earnings inequality.

Figure 4 increases the dispersion of ability, σ_h , holding mean ability at one through $\ln h \sim \mathcal{N}(-\sigma_h^2/2, \sigma_h^2)$. We set $q = 0.06$, $\tau^P = \tau^T = 0$, $\Xi = \xi = 0.5$, and $\epsilon = 7.65$, and adjust τ^L to finance spending of 15% of output.

A more unequal ability distribution increases informality by placing more households at low productivity. This distributional effect dominates the change in relative wages. Informal work also smooths the earnings distribution: in the simulation, increasing the ability Gini from 0.30 to 0.80 raises the earnings Gini from about 0.50 to 0.70. These results are consistent with the positive association between inequality and informality documented in the literature (Chong and Gradstein, 2007; Rosser et al., 2000); Appendix B reports cross-country correlations.

3.3 Welfare Implications

Fiscal policy changes consumption inequality directly through taxes and indirectly through labor reallocation and equilibrium wages. We compute equilibrium for different values of progressivity



Notes: Welfare is population-average expected utility, including labor disutility, in levels. Transfers are zero, and the tax scale adjusts to finance government spending of 15% of output. The two economies in panel (c) share the same ability distribution and fiscal rule; the comparison removes both unemployment risk and informal work.

Figure 5: Progressivity, welfare, and informality

to assess the balance of these effects. We set unemployment risk to $q = 0.06$, the elasticity of substitution to $\epsilon = 7.65$ (Leyva and Urrutia, 2020), informal productivity parameters to $\Xi = \xi = 0.5$, and log-ability dispersion to $\sigma_h = 1$. As above, τ^L adjusts to finance spending of 15% of output without lump-sum transfers. With no progressivity, the informality rate is 62.6% and the earnings Gini is 0.54.

We evaluate welfare using households' expected utility, including the disutility of total labor. Formalization contributes to welfare through its effects on earnings and consumption across households and employment states; the formal share of work is not itself an objective of the welfare calculation.

Figure 5 shows hump-shaped welfare and U-shaped informality. Starting from low progressivity, negative income taxes induce low-ability households to allocate more time to formal jobs. This

inclusion effect outweighs the reduction in formal labor among high-ability households. Lower informality, higher formal earnings, and greater compression of the earnings distribution raise welfare. Redistribution and formalization thus move together over this range of the tax schedule.

Beyond a threshold, the pattern reverses. High marginal rates induce high-ability households to shift more labor into informal work. The evasion margin dominates, shrinking the formal tax base and aggregate output. Welfare peaks at moderate progressivity, where the inclusion gains remain large relative to these costs. In this simulation, the welfare-maximizing degree of progressivity is $\tau^P = 0.24$. Most labor adjustment occurs across sectors; total labor supply changes relatively little, as shown in Appendix E.

Panel (c) compares welfare with an economy without unemployment risk or informality, as in Ferriere et al. (2023). Under the same ability distribution, optimal progressivity in that benchmark is about 0.60, substantially above 0.24 in the model with informality. This comparison changes both the availability of informal work and exposure to unemployment risk. It illustrates how the environment in which redistribution takes place matters: when households can shift labor into untaxed work, high marginal tax rates erode the tax base through an additional channel.

Appendix D.8 varies unemployment risk and public financing needs; both lower optimal progressivity when increased. These results show that the gains from progressive taxation depend on the outside option available to households and on the revenue the government must raise.

The distinction between progressivity and the tax level is central. A higher proportional labor tax lowers the return to formal work across the distribution. Progressivity instead raises formal returns at the bottom while lowering them at the top, allowing redistribution to encourage formality over part of the policy range. The quantitative model asks how far this mechanism operates in Mexico and whether changes in labor or capital tax levels can achieve the same combination of redistribution and formalization.

4 Quantitative Heterogeneous-Agent Model with Informality

The quantitative model preserves the central mechanism of the static model: labor-income taxes are levied on formal earnings only, while households allocate labor between formal and informal activities. Informality therefore serves as an always-available outside option and a margin of tax avoidance. It provides income when formal employment is lost, while earnings in both sectors remain exposed to idiosyncratic productivity risk. The quantitative framework then embeds an

implementable progressive tax schedule into a unified general-equilibrium setting with three key elements to discipline labor-market dynamics, savings behavior, and inequality: (i) an intertemporal consumption–savings problem, (ii) search-and-matching frictions in the formal labor market that make unemployment risk endogenous and heterogeneous, and (iii) a production side with capital, equilibrium prices, and a richer fiscal system.

The static model combines formal-job risk and labor allocation in one decision: households allocate time before learning whether formal work is available. This formulation makes the insurance role of informal work transparent, because its earnings are available whether or not a formal job is found. The quantitative model separates access from hours. Skill-dependent job-finding and separation rates determine access to formal jobs. Once employment status is known, households with a formal job choose how much labor to supply in each sector; those without one can work informally. We denote total hours by ℓ and the formal share by κ , corresponding to n and ϖ in the static model. Households also save to insure against earnings risk, while vacancy posting affects the aggregate rate at which formal jobs become available.

4.1 Taxes and Transfers

The government levies a consumption tax τ^c , a proportional capital-income tax τ^k , a flat tax τ^{profit} on the profit income of firm owners, and a progressive labor-income tax on formal earnings with the same parametric form introduced in Section 3, specified in earnings relative to average gross formal earnings in the benchmark equilibrium, $\bar{y}$:

$$T(y) = y - \tau^L (y/\bar{y})^{1-\tau^P} \bar{y}. \quad (8)$$

Specifying the schedule in relative earnings makes it scale invariant: this allows to anchor each reform at the benchmark level of average formal earnings.⁶ Tax revenues are used to finance government consumption G and lump-sum transfers, τ^T , equal for all households. The government balances its budget every period.

⁶In our calibration units are chosen so that average gross formal earnings equal one at the benchmark, so the schedule coincides with the familiar level form $T(y) = y - \tau^L y^{1-\tau^P}$ at the benchmark and, with the anchor held fixed, in the policy experiments. Writing the schedule in relative earnings is also what makes estimated (τ^L, τ^P) pairs comparable across countries and over time.

4.2 Production

We follow [Leyva and Urrutia \(2020\)](#) and model production as the combination of three technologies. Final output is produced using capital and intermediate inputs, subject to limited managerial capacity (a “span-of-control” constraint) that generates decreasing returns at the firm level and thus positive profits. Intermediate inputs are produced by combining formal and informal labor outputs under imperfect substitutability. In both sectors, labor outputs are produced using linear technologies in labor only. This preserves tractable aggregation and implies an aggregate production function in which measured TFP depends on the share of formal labor in total labor supply.

4.2.1 Final Good Production

A representative firm combines capital K and intermediate inputs H with a Cobb-Douglas technology that features decreasing returns to scale :

$$Y = A \left(K^\alpha H^{1-\alpha} \right)^\nu, \quad (\alpha, \nu) \in (0, 1)^2, \quad (9)$$

with income shares $\alpha\nu$ and $(1 - \alpha)\nu$ for capital and the intermediate composite respectively. The parameter ν captures the span-of-control limitation: when $\nu < 1$, production exhibits decreasing returns at the firm level, generating positive profits that are distributed to household owners.

Let p^f denote the price of formal intermediate output, w^s the return per efficiency unit of informal labor, and w^f the wage per efficiency unit of formal labor. Competitive factor prices equal marginal products. Appendix [G.1](#) gives the factor-price equations and the aggregate production function.

4.2.2 Intermediate Goods Production

The intermediate input is produced by combining formal and informal labor outputs under imperfect substitutability. Formal and informal labor inputs, measured in efficiency units, are denoted by L^f and L^s , respectively. In the formal sector, output is linear in labor with productivity Ω , while in the informal sector output is linear in labor with productivity $\varkappa$. The intermediate composite is therefore given by a CES aggregator with elasticity of substitution ε :

$$H = \left[(\Omega L^f)^{\frac{\varepsilon-1}{\varepsilon}} + (\varkappa L^s)^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}. \quad (10)$$

Reallocating labor between the two technologies changes measured TFP through the composition of production. Over the relevant range, a larger informal share in efficiency units, $\lambda^s = L^s / (L^f + L^s)$, lowers measured TFP; the elasticity of substitution ε governs the size of this effect. Appendix G.1 gives the expression, and Section 5 reports its magnitude in the policy experiments.

4.3 Labor Market

The labor market is segmented into formal and informal sectors. The formal sector is characterized by search and matching frictions, while the informal sector is assumed to be perfectly competitive. Households' formal employment status $F \in \{0, 1\}$ evolves through job-finding and separation, making unemployment risk endogenous, and heterogeneous across workers. As in the static model, households can always choose to supply labor in the informal sector, so informality acts as an outside option that provides income when formal employment is not available and also allows tax avoidance. The model allows for endogenous labor supply across both sectors, and earnings heterogeneity arising from productivity differences, sectoral allocation, and the nonlinear structure of taxation.

Formal Labor Market. The formal labor market is characterized by search and matching frictions. Workers without a formal job ($F = 0$) search for formal employment while having access to informal work as an outside option. A constant-returns-to-scale matching function determines the number of matches, the job-finding rate for workers, and the job-filling rate for firms:

$$M(I, V) = \chi I^\psi V^{1-\psi}, \quad (11)$$

$$f(\theta) = \frac{M}{I} = \chi \theta^{1-\psi}, \quad (12)$$

$$q^v(\theta) = \frac{M}{V} = \chi \theta^{-\psi}, \quad (13)$$

where $\theta \equiv \frac{V}{I}$ is labor market tightness, I is the mass of searching workers (i.e., households with $F = 0$), and V is the number of vacancies posted in the formal sector.

4.4 Households

There is a continuum of infinitely lived households deriving utility from consumption and leisure. Households are endowed with one unit of time, which they can allocate to formal or informal labor supply (or both). They can save in a risk-free asset, which pays an equilibrium interest

rate r , constant within each steady state and subject to a capital income tax τ^k . They pay a proportional tax on consumption expenditures τ^c and a progressive tax on formal labor income (8). The government also uses a uniform lump-sum instrument τ^T , which is a transfer when positive and a levy when negative. The household's problem is to choose consumption, savings, and labor supply across sectors in order to maximize expected lifetime utility.

Idiosyncratic Risk. Households face idiosyncratic risk in labor productivity, formal employment attachment, and ownership status. Labor productivity evolves stochastically according to a log-AR(1) process:

$$\ln h' = \rho \ln h + \sigma_h \epsilon_h, \quad \epsilon_h \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu_h, 1). \quad (14)$$

with persistence ρ and innovation standard deviation σ_h .⁷

Sector-Specific Worker Productivity. Formal and informal sectors differ in technology, captured by the sectoral productivity factors Ω and $\varkappa$ (see Section 4.2.2). At the worker level, we adopt the same sector-specific productivity mapping as in the static framework: effective productivity in the formal sector equals potential productivity, $h^f = h$, while effective productivity in the informal sector is $h^s = \Xi h^\xi$. The parameters Ξ and ξ govern the level and curvature of informal-sector productivity. We set $\xi \in [0, 1]$; values below one compress proportional skill differences in informal efficiency units.

We interpret $\xi < 1$ as a reduced form for limits on scaling informal activity, including restricted access to capital, customers, and enforceable contracts. The panel evidence in Section 2 supports lower informal returns to skill, with an estimate around 0.8. The baseline calibrates ξ jointly with the other informal-productivity parameters using sectoral earnings moments;; Appendix C also considers a calibration that fixes ξ at the panel estimate.

Risk across sectors. A single persistent skill shock affects both sectors through $h^f = h$ and $h^s = \Xi h^\xi$. Informal productivity responds with elasticity ξ , so its proportional response to the common shock is smaller when $\xi < 1$. With the calibrated ξ close to one, this effect is small. Formal employment also faces skill-specific job-finding and separation rates, while informal work can be taken up immediately and at any intensity. Informal work therefore provides income after a

⁷In line with the static model, innovation mean parameter μ_h is set to satisfy the steady-state normalization condition $\mathbb{E}[h] = 1$.

formal-job loss while remaining exposed to productivity risk.⁸ Lower informal returns to skill also raise the earnings cost of evasion for high-productivity households. These returns and formal-job risk jointly shape the allocation of labor across sectors.⁹

Skill-Dependent Transition Rates. The formal employment status is a binary variable indicating whether the household is employed in the formal sector ($F = 1$) or not ($F = 0$). The average job-finding rate into the formal sector depends endogenously on labor market tightness, $\mathbb{E}(f) = f(\theta)$, while the average separation rate $\bar{\iota}$ is exogenous. To capture the systematic heterogeneity in transition probabilities observed across the labor-income distribution, we assign productivity-specific job-finding and separation rates directly over the h -productivity grid.

Formally, each productivity type $h_i \in \{h_1, \dots, h_{n_h}\}$ is associated with a pair of transition parameters $\{\Delta f_{h_i}, \iota_{h_i}\}$, where Δf_{h_i} captures the deviation around the endogenous equilibrium job-finding rate $f(\theta)$, and ι_{h_i} denotes the exogenous separation rate. The arrays $\{\Delta f_{h_i}\}$ and $\{\iota_{h_i}\}$ are calibrated to match the empirical profiles of job-finding and separation rates across labor-income deciles, ensuring that heterogeneity in labor market dynamics is consistent with the data.¹⁰

Employment status is known when households choose hours and savings. Thus, F determines access to a formal job, while the formal labor share κ determines how much labor a household with access supplies formally. Households with $F = 0$ can work only informally.

Risky Firm-Ownership Status. We assume that only workers with the discretized highest labor productivity level who are attached to the formal sector can access the ownership status with some probability π^{up} . Following [Castaneda et al. \(2003\)](#), we assume this “awesome” status has a shorter expected duration relative to other idiosyncratic states pinned down by the probability of losing this status π^{fall} . Conditional on falling, the probability of ending up in any of the discretized

⁸This distinction is consistent with longitudinal evidence from Brazil. [Gomes et al. \(2020\)](#) document greater earnings volatility among informal workers and large earnings changes around formal–informal transitions. [Engbom et al. \(2022\)](#) similarly find that earnings changes are substantially more dispersed among informal than formal workers, with particularly high volatility among sector switchers. Informal work can therefore provide income after formal employment is lost while exposing workers to substantial earnings fluctuations. These findings are also consistent with our Mexico-specific ENOE evidence in Section 2.3, where the estimated transitory component of earnings variance is 2.4 times larger in informal employment.

⁹Informal jobs are not subject to destruction shocks in the model; informal workers face risk through stochastic efficiency units. Empirically, informal labor markets display larger worker flows than formal ones ([Donovan et al., 2023](#); [Meghir et al., 2015](#)). The baseline also abstracts from transitory earnings risk. Appendix C reports a calibration with sector-specific transitory risk estimated from the ENOE panel.

¹⁰Worker productivity is discretized over a grid of length n_h using importance sampling, which enables the model to replicate observed separation rate profiles across labor–income deciles by projecting the data-decile rates onto the discretized grid. Idiosyncratic transition rates satisfy: $\mathbb{E} \Delta f_h = 0$ and $\mathbb{E} \iota_h = \bar{\iota}$

productivity level follows an exponential distribution with rate π^{unlucky} .¹¹

4.4.1 Informally Employed Households' Problem

Self-employed households are assumed to be informal workers, who supply labor in the informal sector. They supply their labor ℓ in a perfectly competitive market, where they receive the informal wage w^s per efficiency unit of labor. The self-employed households' problem is given by:

$$\begin{aligned}
V^s(a, h) &= \max_{a', \ell} \left\{ u(c, 1 - \ell) + \beta \mathbb{E}_h \left[f_h V^f(a', h') + (1 - f_h) V^s(a', h') \right] \right\}, \\
\text{s.t.} \quad & (1 + \tau^c)c + a' = \underbrace{\left[1 + (1 - \tau^k)r \right]}_{\equiv \tilde{R}} a + \ell w^s h^s + \tau^T, \\
& \ln h' = \rho \ln h + \sigma_h \epsilon'_h, \quad \epsilon'_h \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu_h, 1), \\
& c \geq 0, \quad a' \geq 0, \quad \ell \in [0, 1].
\end{aligned} \tag{15}$$

where V^s is the value function of self-employed households, V^f is the value function of formally employed households, and $\mathbb{E}_h$ is the conditional expectation operator given household's current potential productivity level h .

Let u_c denote marginal utility of consumption and u_2 marginal utility of leisure, the second argument of $u(c, 1 - \ell)$. At an interior solution, the self-employed household's labor choice satisfies: $\frac{w^s h^s}{1 + \tau^c} \cdot u_c = u_2$.

Saving. When the borrowing constraint does not bind, the intertemporal condition in either employment state is the standard Euler equation: $u_c = \beta \tilde{R} \mathbb{E}_{h, F} u'_c$. The equality becomes an inequality when the borrowing constraint binds.

4.4.2 Formally Employed Households' Problem

Formally employed households can supply labor in both sectors choosing the share of labor allocated to formal employment κ and the total labor supply ℓ . The formal-informal trade-off is generated by progressive labor taxation in the formal sector and productivity differentials across the two sectors.

¹¹The ownership state is a parsimonious device to match top-end wealth concentration; entry and exit probabilities do not respond directly to taxes. Profit income $\Pi(h)$ is taxed at the flat rate $\tau^{\text{profit}} = 21.1\%$, the effective business income taxation for Mexico estimated by [Bachas et al. \(2023\)](#), with the revenue accruing to income-tax collections.

Let $\tilde{R} = 1 + (1 - \tau^k)r$ denote the gross after-tax asset return. Their problem is:

$$\begin{aligned}
V^f(a, h) &= \max_{a', \ell, \kappa} \left\{ u(c, 1 - \ell) + \beta \mathbb{E}_h \left[\iota_h V^s(a', h') + (1 - \iota_h) V^f(a', h') \right] \right\}, \\
\text{s.t.} \quad & (1 + \tau^c)c + a' = \tilde{R}a + y^{\text{net}}(\kappa, \ell, h) + \tau^T + (1 - \tau^{\text{profit}}) \Pi(h), \\
& y^{\text{net}}(\kappa, \ell, h) = (1 - \kappa) \ell w^s h^s + \tau^L \bar{y}^{\tau^P} \left(\kappa \ell w^f h^f \right)^{1 - \tau^P}, \\
& \ln h' = \rho \ln h + \sigma_h \epsilon'_h, \quad \epsilon'_h \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu_h, 1), \\
& c \geq 0, \quad a' \geq 0, \quad (\ell, \kappa) \in [0, 1]^2.
\end{aligned} \tag{16}$$

The expectation $\mathbb{E}_h$ is conditional on current productivity h . At an interior solution, formally employed households' labor supply and allocation across sectors jointly satisfy:

$$\begin{aligned}
u_2 &= \frac{(1 - \kappa) w^s h^s + \tau^L \bar{y}^{\tau^P} (1 - \tau^P) \ell^{-\tau^P} (\kappa w^f h^f)^{1 - \tau^P}}{1 + \tau^c} \cdot u_c, \\
\ell w^s h^s &= \tau^L \bar{y}^{\tau^P} (1 - \tau^P) \kappa^{-\tau^P} (w^f \ell h^f)^{1 - \tau^P}.
\end{aligned} \tag{17}$$

The second condition equates the marginal after-tax return to formal work with the return to informal work. Taxes can therefore change sectoral labor supply even when a household's formal-employment status does not change.

4.5 Formal Labor Agency

Formal intermediate goods are contracted through a representative labor agency that posts vacancies to hire formal workers in a frictional labor market. The labor agency is owned by the households in the ownership state, who receive its net surplus, and its revenues depend on the effective units of formal labor supplied: $L^f = \int \underbrace{\kappa \ell h}_{\equiv \ell^f} d\mu(a, h, F)$. The labor agency recursive problem is given by:

$$\begin{aligned}
J(L^f) &= \max_V -c_v V + (\Omega p^f - w^f) L^f + \tilde{R}^{-1} J(L^{f'}) \\
\text{s.t.} \quad & L^{f'} = \int [(1 - \iota_h) \mathbf{1}_{\{F=1\}} + f_h \mathbf{1}_{\{F=0\}}] \ell^{f'} d\mu(a, h, F),
\end{aligned} \tag{18}$$

where J is the value of the labor agency, c_v is the cost of posting a vacancy V , and $\{\iota_h, f_h\}$ are the productivity-specific separation and job-finding rates, respectively. The agency discounts future payoffs at the gross after-tax asset return $\tilde{R}$. The steady state equilibrium condition is given by the

zero-profit condition for the labor agency in a stationary labor market, i.e. $L^f = L^{f'}$:

$$c_v = \tilde{R}^{-1} \frac{(\Omega p^f - w^f) L^f}{V} \quad (19)$$

The zero-profit condition (19) implies that the cost of posting one vacancy c_v is equal to the expected average revenue from hiring a formal worker. Vacancy posting changes labor-market tightness and hence the stationary distribution of households with formal jobs.

The labor agency receives p^f per unit of formal output and pays w^f per efficiency unit of formal labor. Due to search frictions, there exists a range of wages between the reservation values of households and the labor agency. With Nash bargaining, household-specific outside options would make wages depend on both wealth and productivity. We instead use a wage rule that splits the flow surplus of a formal match, $\Omega p^f - w^s$, between the household and the labor agency:¹²

$$w^f = w^s + \eta (\Omega p^f - w^s). \quad (20)$$

These household responses feed back into vacancy posting and job finding through the surplus earned on formal labor. The wage-rule parameter η governs how this surplus is split between households and the labor agency. Moving η by ± 0.15 around its baseline value of 0.60 shifts optimal progressivity by +0.01 and -0.005 , respectively (Appendix D.5).

4.6 Equilibrium

A stationary equilibrium combines optimal household choices, competitive factor prices, vacancy posting, and an invariant distribution $\mu(a, h, F)$. Household asset holdings finance capital, and labor supplied in each sector equals labor demand. Let g^a , g^ℓ , g^κ , and g^c denote household saving, hours, sectoral allocation, and consumption policies. The government balances its budget:

$$\begin{aligned} G + \tau^T = \int \left\{ T \left(g^\kappa(a, h, F) g^\ell(a, h, F) w^f h^f \right) + \tau^c g^c(a, h, F) \right. \\ \left. + \tau^k r \cdot a + \tau^{\text{profit}} \Pi(h) \right\} d\mu(a, h, F). \end{aligned} \quad (21)$$

In the benchmark and the main policy experiments, $G/Y = 15\%$ and the lump-sum instrument τ^T adjusts. Appendix G.2 gives the full equilibrium definition and market-clearing conditions; Appendix D.2 considers alternative budget rules.

¹²Similar wage rules are used in [Gornemann et al. \(2016\)](#), [Den Haan et al. \(2018\)](#), and [Graves \(2025\)](#).

4.7 Calibration

Existing reform evidence does not provide direct estimates of the formal–informal labor-supply elasticities relevant for our policy analysis. We therefore discipline these margins through a joint calibration to Mexican evidence on sectoral earnings, employment transitions, and the distributions of wealth and consumption.

Households have additively separable preferences over consumption and leisure with constant relative risk aversion, following the [King et al. \(1988\)](#) paradigm, which allows for wealth effects in labor supply decisions. Flow utility is given by

$$u(c, 1 - \ell) = \frac{c^{1-\zeta} - 1}{1 - \zeta} + \Phi \frac{(1 - \ell)^{1-\phi} - 1}{1 - \phi}. \quad (22)$$

The model is calibrated using a simulated method of moments to match aggregate and cross-sectional moments for Mexico in 2024. Calibration targets are grouped into three blocks: (i) inequality and redistribution; (ii) labor market outcomes; and (iii) fiscal variables. Table 2 reports the targeted moments and their associated parameters, while Table 3 summarizes the remaining parameters. These pairings summarize how the moments inform the calibration. Each parameter can affect several moments through household choices and equilibrium prices.

Inequality and Redistribution. The first group of moments disciplines the cross-sectional distribution of wealth. The capital-to-output ratio disciplines the discount factor. To reproduce the observed concentration of wealth at the top of the distribution, the model features a stochastic high-income ownership state following [Castaneda et al. \(2003\)](#). Entry into this state is restricted to top-productivity formal workers and occurs at a constant rate, calibrated so that owners are 0.62% of households, the share of formal employers with six or more workers among the employed in the ENOE (2024). The Gini coefficient of wealth in turn disciplines σ_{LR} , the long-run dispersion of the persistent efficiency-unit process, which drives wealth concentration through the bulk of the distribution.

Earnings inequality is disciplined using moments that characterize earnings levels and dispersion across sectors. Parameters governing informal-sector productivity and the mapping of idiosyncratic productivity across sectors are chosen to match the formal–informal gap in mean log earnings and the persistent component of the informal–formal gap in log-earnings variance. These moments allow

Table 2: Calibration targets and parameter values

Target	Data	Model	Parameter	Description	Value
K/Y	2.44	2.46	β	discount factor	0.92
Gini wealth	0.72	0.76	σ_{LR}	long-run s.d. of log efficiency units	1.00
owner share of households (%)	0.62	0.62	π^{up}	ownership entry	0.022
mean log-earnings gap	0.72	0.71	$\varkappa$	informal linear productivity	1.87
log-earnings variance gap, persistent	0.17	0.17	ξ	informal skills curvature	0.92
informal/formal net wage ratio	0.92	0.90	Ξ	informal skills scaling	0.81
Gini consumption	0.41	0.41	τ^P	earnings tax progressivity	0.015
L (hours / time endowment)	0.414	0.414	Φ	leisure weight	0.33
informality rate (headcount)	0.544	0.564	c_v	vacancy cost	0.015
avg. job-finding rate	0.159	0.168	χ	matching function TFP	0.159
separation rates	see Fig. 6a		ι_h	h -specific sep. rates	see ^a
job-finding rates	see Fig. 6b		Δf_h	h -specific job-f. rate diffs	see ^a
consumption tax revenues / Y	0.056	0.057	τ^c	consumption tax rate	0.083
income tax revenues / Y	0.074	0.073	τ^k	capital tax rate	0.081
tax rate at mean formal earnings	0.128	0.125	τ^L	labor-tax scale	0.875

Notes: Baseline calibration for Mexico. Parameter–moment pairings summarize the joint calibration. K is capital and Y is output. The owner share is the share of formal employers with six or more workers among the employed. The mean log-earnings gap is formal minus informal; the variance gap is the persistent component of the informal-minus-formal gap in the variance of log earnings (Section 2.3). The wage ratio is the informal to after-tax formal hourly wage at mean formal earnings, from the salaried lag-matched joint estimate of Section 2.3. L is average weekly hours worked over a 100-hour endowment. Informality is the share of employed workers in informal jobs; in the model, a worker is informal if less than half of her hours are formal. The ownership entry probability is the effective per-period probability. Income-tax revenue includes labor, capital, and profit taxes. The last row reports the average tax rate at mean formal earnings. Labor-market targets use the National Survey of Occupation and Employment (ENOE, 2024); fiscal targets use OECD Taxing Wages (OECD, 2026) and OECD Revenue Statistics (OECD, 2025). The capital-output ratio is the 2023 net capital stock per unit of GDP from the European Commission’s AMECO database; the wealth Gini is from UBS (2025); the consumption Gini is from ENIGH 2024.

^a Transition rates are annual flows between informal and formal employment by labor-income decile, assigned across productivity types using the profiles in Figure 6.

the model to reproduce the empirical formal–informal earnings patterns documented in our data, with informal earnings exhibiting lower average levels but higher dispersion than formal earnings.

Three parameters govern informal productivity and are calibrated jointly with the three sectoral-earnings moments: the linear productivity of the informal technology, $\varkappa = 1.87$, and the level and curvature of the mapping from ability to informal efficiency units, $\Xi = 0.81$ and $\xi = 0.92$. The formal-sector productivity scale is instead a normalization of units: Ω is set to 2.12, the value at which average gross formal earnings per full-time-equivalent formal worker equal one at the benchmark, so that the tax schedule takes its familiar level form there. The informal technology is thus about 12% less productive than the formal one ($\varkappa/\Omega = 0.88$), before accounting for the lower returns to ability in informal work.

We choose tax progressivity to match consumption inequality, conditional on the model’s wealth and earnings distributions. The tax schedule changes how earnings shocks translate into disposable income and consumption, thereby affecting households’ remaining exposure to risk. Panel 6d validates the calibrated schedule: the effective tax rates it implies are close to those in Mexican fiscal data.

Two targeted moments summarize sectoral earnings levels and dispersion. In the data, the gap in mean log earnings between formal and informal workers is 0.72, and the persistent component of the variance of log earnings is 0.17 *higher* among informal workers (Section 2.3). Because the baseline income process has no transitory component, we target the persistent part of the variance gap. The calibrated model reproduces both moments, as informal work is worse paid and more dispersed than formal work: the mean gap is 0.71 and the variance gap 0.17. With a single persistent skill process common to both sectors and informal returns a concave function of skill ($\xi < 1$), the informal sector *compresses* skill dispersion, so the model generates the informal variance excess through sorting and composition.¹³

Labor Market Outcomes. The calibration of the labor market block targets moments describing the allocation of workers across sectors and the flows between employment states. Parameters governing labor supply and search frictions are chosen to match average hours worked, the informality rate, the average job-finding rate in the formal sector, and the job-finding and separation rates by labor-income decile. Average hours are 41.4 per week in the ENOE (2024), 0.414 of a 100-hour

¹³Appendix C reports two robustness calibrations: one adds the sector-specific transitory earnings risk measured in the panel and targets the full variance gap; the other fixes ξ at its panel estimate, with the panel’s earnings persistence. In both, the welfare-maximizing progressivity remains several times its benchmark level.

weekly endowment, and the leisure weight Φ matches them exactly given the leisure curvature. The informality rate is the share of employed workers in informal jobs, 0.544; in the model, a worker is informal if less than half of her hours are formal. The transition rates are annual flows between informal and formal employment among workers employed at two ENOE interviews one year apart (2023 to 2024), by labor-income decile; on average, 15.9% of informal workers find a formal job and 16.9% of formal workers move to informality within a year. These moments are informative about the weight of leisure in utility, the cost of posting vacancies, and the productivity-specific slopes of job-finding and separation schedules.

Fiscal Tools. The labor-tax scale τ^L in (8) is calibrated using the observed average tax rate at mean formal earnings of 12.77%, consistent with [OECD \(2026\)](#). The consumption tax τ^c targets consumption-tax revenue, while the capital-income tax τ^k targets income-tax revenue, both relative to GDP and averaged over 2021–24, from the OECD Revenue Statistics database ([OECD, 2025](#)). Progressivity τ^P targets the consumption Gini, while the lump-sum instrument τ^T balances the government budget. Equilibrium government expenditure is set at 15% of GDP, financed through tax revenues from consumption (5.7 pp) and household income (labor, capital, and business profits; 7.3 pp). The remaining share is financed by a uniform lump-sum levy of about 2 percent of GDP, which stands in for Mexico’s substantial non-tax revenues: total public expenditure in Mexico is around 25% of GDP, with roughly 10 percentage points financed through oil revenues, other non-tax revenues, and fiscal deficits rather than through the tax instruments in the model. In the policy experiments this residual instrument rebates or collects revenue changes uniformly. Its level affects household resources and hence labor supply and saving, even though it creates no direct marginal tax wedge.

Remaining Parameters. Table 3 reports the remaining parameters. Relative risk aversion is 2 and the leisure curvature ϕ is set so that the Frisch elasticity at average hours is 0.3, following [Leyva and Urrutia \(2020\)](#). The capital exponent within the input composite, α , is 0.23, annual depreciation is 0.06, and the elasticity of substitution between formal and informal inputs is 7.70. The matching elasticity is 0.6, and the wage-rule parameter η is set to the same value.

The baseline income process has annual persistence $\rho = 0.98$. The stationary productivity distribution alone does not identify persistence. Persistence nevertheless affects wealth and consumption through saving and self-insurance. Appendix C reports the fit and policy implications of adding sector-specific transitory risk and of fixing ξ at its panel estimate together with a lower earnings

Table 3: Remaining parameters for the baseline economy

Parameter	Symbol	Value	Parameter	Symbol	Value
Risk aversion	ζ	2.00	Income persistence	ρ	0.98
Frisch elasticity	$\frac{1-\ell}{\ell}\phi^{-1}$	0.30	Capital depreciation	δ	0.06
Formal wage rule	η	0.60	Matching function exp.	ψ	0.60
Span-of-control (DRS)	ν	0.96	Capital exponent in composite	α	0.23
Profit tax rate	τ^{profit}	0.211	Fall-distribution rate	π^{unlucky}	2.00
Formal productivity (normalization)	Ω	2.12	Ownership exit rate	π^{fall}	0.21
Formal–informal elast. of substitution	ε	7.70			

Notes: Depreciation and income persistence are annual. Preferences follow [Leyva and Urrutia \(2020\)](#); the Frisch elasticity is evaluated at average hours, $\ell = 0.414$, which gives $\phi = 4.72$. π^{up} (Table 2) and π^{fall} govern entry and exit; π^{unlucky} is the exponential rate governing productivity after exit. The exit probability is the effective per-period probability; the entry probability is calibrated to the owner share of households. Formal productivity Ω normalizes average gross formal earnings per full-time-equivalent formal worker to one at the benchmark.

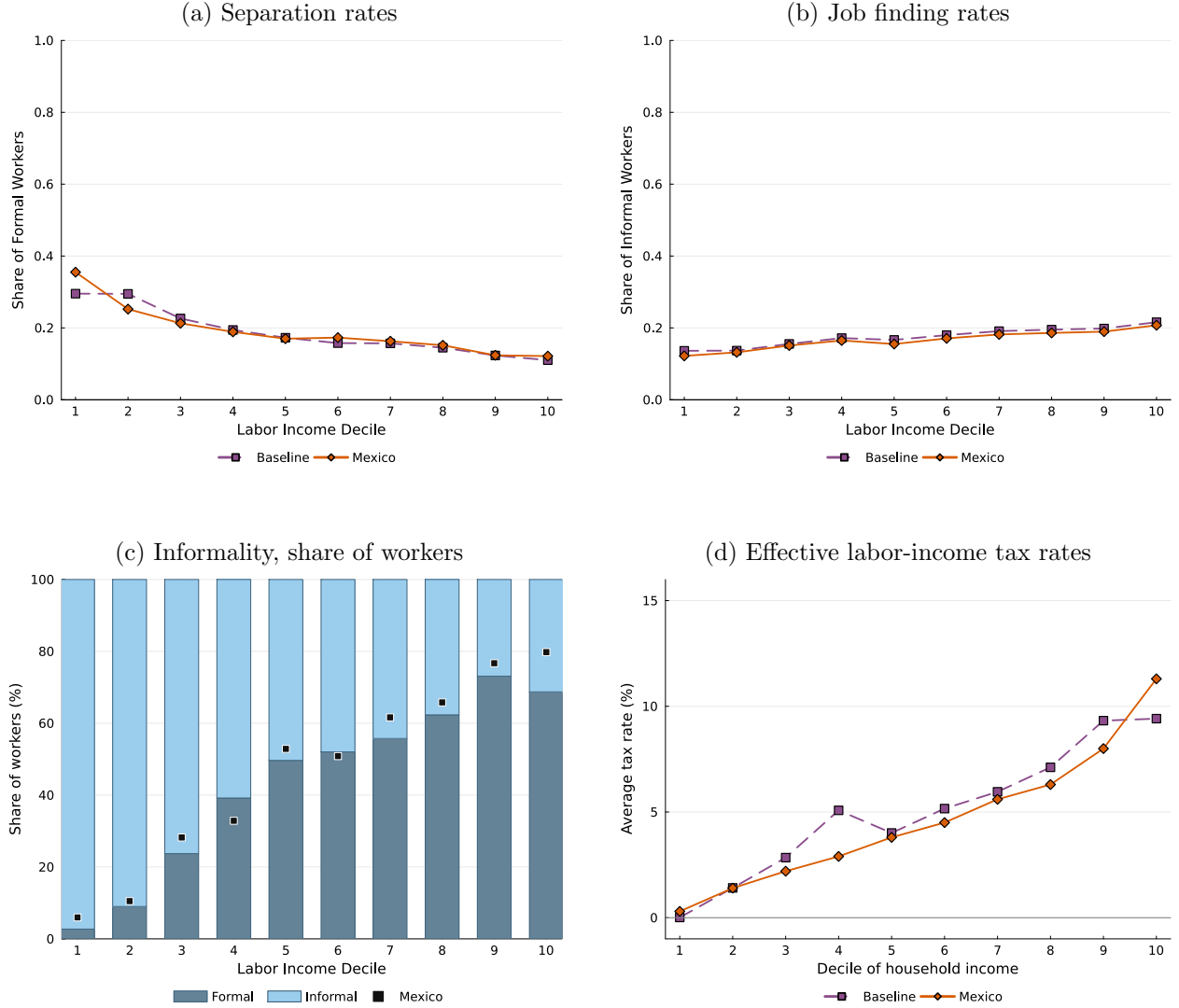
persistence.

4.7.1 Fit of the Baseline Economy

Figure 6 compares the baseline labor-market gradients with ENOE data and the effective labor-income tax rates with Mexican fiscal data from the Secretaría de Hacienda y Crédito Público (SHCP). The separation and job-finding profiles are calibration targets: low-income workers face higher separation risk and lower job-finding rates, which make informal work a more valuable outside option. The informality gradient combines these differences in access to formal jobs with households’ labor-allocation choices.

Formality increases along the decile profile in data and model, both as a share of workers (Panel 6c) and as a share of hours worked (Appendix Panel D.4a), with substantial informal work remaining at the top. This pattern does not require heterogeneous transition rates: Appendix D.6 reports an economy with common job-finding and separation rates in which the gradient remains downward sloping. In the aggregate, the informal share of hours worked is 0.581 in the model and 0.514 in the data. Within the informal sector, the ratio of the 20th to the 80th percentile of earnings is 0.20 in the model and 0.30 in the data.

Panel 6d validates the benchmark tax schedule against Mexican fiscal data that the calibration does not target. It compares the model’s labor-income tax as a share of total gross household income, by decile of household income, with the effective rate of the income tax on salaries that



Notes: In Panels 6a–6c, data are from Mexico’s National Survey of Occupation and Employment (ENOE, 2024), and workers are ranked by labor income. Panels 6a and 6b show transition profiles targeted in the baseline calibration: annual flows between informal and formal employment by labor-income decile. Panel 6c shows the shares of formal and informal workers in the model (bars) and the formal share in the data (squares); in the model, a worker is informal if less than half of her hours are formal. Panel 6d is not targeted. It shows the model’s labor-income tax paid over total gross household income (labor and capital income), by decile of total gross income, excluding entrepreneurs, and the income tax on salaries over total gross household income, by decile of households, as estimated by the Secretaría de Hacienda y Crédito Público (SHCP) from ENIGH 2024 in the scenario that assigns the tax only to earners affiliated with social security (*Distribución del pago de impuestos y recepción del gasto público por deciles de hogares y personas*, 2026). Appendix D.7 reports the version that includes entrepreneurs and the tax on their profits.

Figure 6: Baseline economy labor-market gradients and effective tax rates

Mexico’s Ministry of Finance (SHCP) estimates from the 2024 household income and expenditure survey (ENIGH). The benchmark progressivity yields effective tax rates by income decile in line with those measured in Mexico.

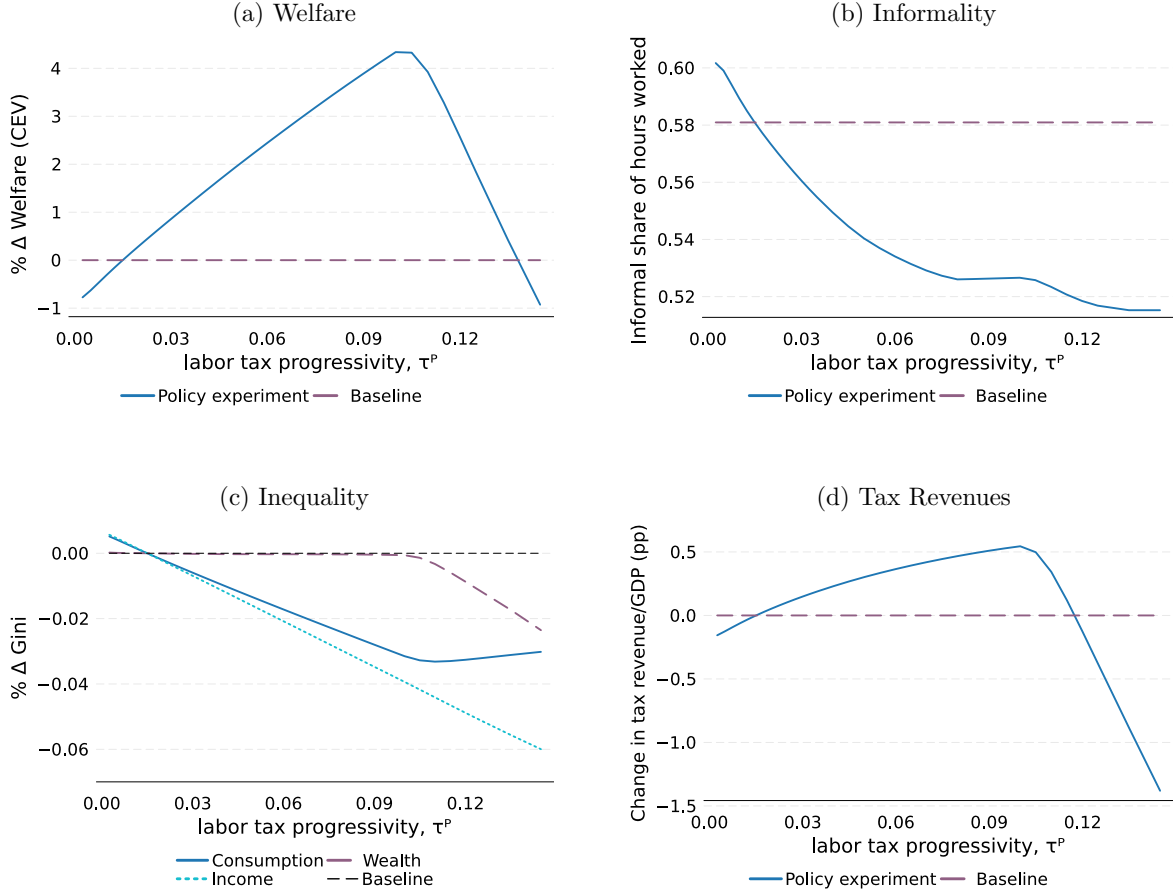
5 Optimal Progressivity with Informality: Quantitative Results

We study how progressive labor taxation changes welfare and informality, and compare it with changes in labor- and capital-tax levels. In the benchmark experiments, government consumption remains at its calibrated share of output, $G/Y = 15\%$. Every reform balances the government budget: the lump-sum instrument rebates or collects changes in revenue net of spending. We maximize utilitarian welfare over each instrument’s explored range, varying one tax-rate parameter at a time while holding the others and structural parameters fixed. Prices, household choices, and the stationary distribution adjust. Welfare includes consumption and leisure and is evaluated over each economy’s stationary distribution. Consumption-equivalent variation (CEV) compares these stationary equilibria with the baseline. Appendix D.4 states the criterion.

5.1 Optimal Progressive Labor Taxation

Figure 7 reports our main experiment. Welfare is hump-shaped in progressivity, peaking at $\tau^P = 0.10$, about six-and-a-half times the benchmark level of 0.015. At the optimum, welfare rises by 4.3% of consumption, informality, measured as the informal share of hours worked, falls by 5.4 percentage points, and consumption inequality falls by 3.2%. Tax revenues increase by about half a percentage point of GDP. Welfare and revenue as a share of GDP peak at the same degree of progressivity. The optimum is the point at which the evasion margin opens. Up to $\tau^P = 0.10$, welfare rises almost linearly with progressivity, while the most productive households with a formal job keep their formal-work hours. Beyond it, these households begin to shift hours into informal activities, and welfare and revenues decline together as the evasion margin erodes the formal tax base. Informality instead falls steeply up to $\tau^P \approx 0.08$, is flat up to the optimum, and keeps declining slowly beyond it, because wider access to formal jobs outweighs the hours that top earners move into informal work.

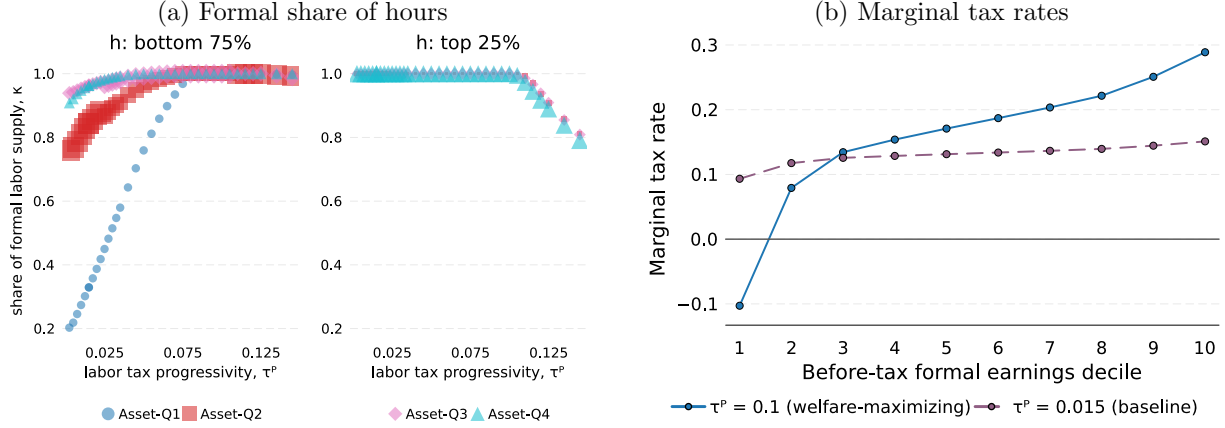
The allocation of formal labor in Panel 8a shows how the two margins operate. Households in the lower part of the productivity distribution, especially those with low assets, expand formal labor supply as progressivity rises. Negative effective tax rates increase the return to formal work



Notes: Labor-tax progressivity τ^P varies from its baseline of 0.015, with other tax and structural parameters fixed. Government consumption remains 15% of output, and the lump-sum transfer balances the budget. Welfare gains are steady-state consumption-equivalent variations (CEV, percent), including leisure, relative to the baseline. Informality is shown in levels. Gini changes are proportional changes from the baseline: -0.01 denotes a 1% decline. Panel (d) reports changes in tax revenues relative to GDP.

Figure 7: Optimal progressivity: welfare, informality, inequality, and revenue

and draw these households into the tax base. By $\tau^P \approx 0.08$ they supply all their hours formally, and the fall in informality in Panel 7b levels off. High-productivity households do not respond up to the optimum, but reduce their formal labor supply beyond it, reallocating effort toward untaxed informal activities. The inclusion margin therefore dominates at low progressivity, while the evasion margin opens at the optimum and becomes stronger as marginal rates rise. Informality nonetheless keeps falling beyond the optimum, because the evasion margin also widens access to formal jobs: as top earners move hours out of formal work, the labor agency's surplus per unit of formal labor rises and it posts more vacancies. Job finding rises for all workers without formal access, and those who gain it supply most of their hours formally, so wider access lowers the informal share of hours



Notes: Government consumption remains 15% of output, and the lump-sum transfer balances the budget as progressivity varies; other tax parameters are fixed. Panel (a) shows the formal share of hours κ of households with a formal job by asset quartile, separately for the bottom 75% and top 25% of productivity, entrepreneurs included in the top group. Marker area is proportional to the cell's share of all households. Panel (b) compares the baseline and welfare-maximizing marginal tax schedules at mean gross formal earnings within each decile of the baseline distribution of formal earners.

Figure 8: The inclusion and evasion margins

by more than the top earners' reallocation raises it (Appendix D.1).

Panel 8b translates the reform into marginal tax rates along the distribution of gross formal earnings. The baseline schedule is nearly flat, with rates between 9% and 15% across deciles. The welfare-maximizing schedule has a negative marginal rate in the bottom decile, crosses the baseline schedule between the second and third deciles, and reaches 29% in the top decile. Average taxes are negative for roughly the bottom sixth of formal earners, whose earnings are below a quarter of the mean. Higher taxation at the top thus finances subsidies to formal work at the bottom.

The three inequality measures respond differently. At the optimum, labor-income and wealth Ginis are 3.9% and 0.06% below the benchmark, and both continue to fall at higher progressivity. Consumption inequality instead falls until just beyond the optimum and then rises slightly. As the formal base contracts, revenue falls and the lump-sum fiscal instrument collects more from households. Consumption inequality nonetheless stays below its benchmark level up to the top of the grid, where it is 3.0% lower, while income and wealth continue to equalize. Panels 7c and 7d show these distributional and fiscal responses.

Reallocation also changes measured TFP through the composition of formal and informal labor, as described by equation (G.5). The effect is small at the optimum: TFP is 0.004% above its benchmark value. At the top of the progressivity grid, $\tau^p = 0.145$, it is 0.26% lower. The difference

reflects the productivity of the workers who move across sectors. Inclusion draws in low-productivity workers, whereas evasion removes high-productivity workers. The welfare gains at the optimum arise mainly through insurance, redistribution, and the formal tax base, with little change in measured TFP.

The direction of these responses is consistent with evidence on changes in the relative payoff to formal work. More generous U.S. EITC supplements reallocate labor from informal to regular employment (Gunter, 2013), and a temporary wage incentive conditional on formal employment in Mexico increases formal employment partly by reducing informal employment (Abel et al., 2022). Policies that reduce the relative payoff to registered work operate in the opposite direction: an income-tested transfer in Uruguay reduces formal employment (Bérgolo and Cruces, 2021), while Mexico’s Seguro Popular reduces formal registration (Bosch and Campos-Vázquez, 2014).

Evidence on the evasion margin is more limited. Slonimczyk (2012) finds that Russia’s 2001 tax reform reduced informal employment, with stronger responses among individuals receiving larger tax reductions. Using the same reform, Gorodnichenko et al. (2009) document substantial changes in tax evasion associated with voluntary compliance. These reforms provide evidence on the incentive channels through which progressivity operates. Their effects reflect the particular policy changes and populations studied. Our quantitative framework combines household choices with changes in wages and job availability to evaluate reforms of the full progressive tax schedule.

5.2 Progressivity Compared with Other Tax Instruments

Can changes in the level of labor or capital taxation achieve similar welfare gains? We compare the three instruments under the same government budget rule, holding G/Y fixed and returning revenue changes to households. This gives each tax increase a redistributive component. The instruments differ in how they change the return to formal work and the size of the formal tax base.

Table 4 reports the best reform for each instrument over the explored ranges. Raising the labor-tax level $(1 - \tau^L)$ by 10 percentage points delivers a welfare gain of 6.3% of consumption. A 9-percentage-point increase in the capital income tax delivers a gain of 1.6%, with welfare still rising at the upper edge of the grid. Progressivity yields 4.3%. The differences in informality are much larger: it falls by more than five percentage points under higher progressivity, rises by almost twenty points under the higher labor-tax level, and changes little under capital taxation. Progressivity is the only instrument that raises welfare and materially reduces informality.

The higher labor-tax level achieves its larger gain by raising much more revenue, which the bud-

Table 4: Fiscal reforms and the trade-off between welfare and informality

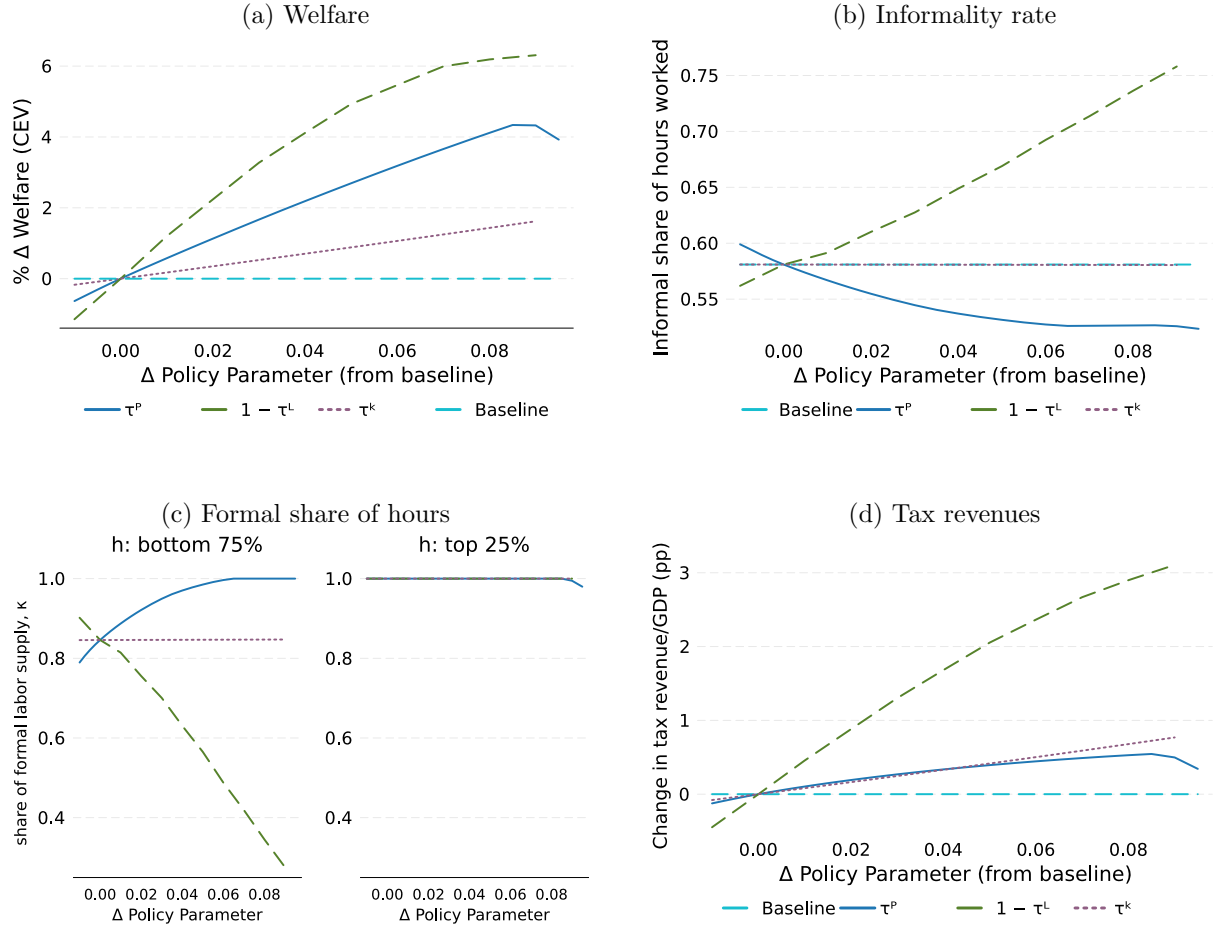
	Reform (pp)	Welfare CEV (%)	Informality change (pp)	Revenue/GDP change (pp)	CEV at equal revenue (%)
<i>G/Y fixed, revenue changes returned to households</i>					
Progressivity, τ^P	+8.5	4.34	-5.43	+0.54	4.34
Labor-tax level, $1 - \tau^L$	+10.0	6.33	+19.65	+3.28	1.42
Capital tax, τ^k	+9.0 [†]	1.62	-0.05	+0.77	1.16
<i>Transfer fixed, government consumption adjusts</i>					
Progressivity, τ^P	+10.0 [‡]	2.65	-6.03	+0.11	2.65
Labor-tax level, $1 - \tau^L$	—	—	—	—	-0.11
Capital tax, τ^k	—	—	—	—	-0.18

Notes: Each row maximizes welfare over the explored range for one instrument, holding other tax parameters fixed. Changes are relative to the baseline. CEV denotes steady-state consumption-equivalent variation, including leisure. Informality is the informal share of hours worked. Government consumption has no direct utility value. The labor-tax level $1 - \tau^L$ is the average tax rate at mean formal earnings. [†] marks a grid boundary, not an interior optimum. In the second block, welfare is maximized subject to government consumption remaining at least at its baseline share of output, 15%; [‡] marks an optimum at which this requirement binds. — marks no reform: the best reform is the baseline itself. The last column reports each instrument’s CEV at the reform that changes revenue/GDP by as much as the progressivity reform in the same block (+0.54 and +0.11 percentage points). The grids are $\tau^P \in [0.0025, 0.145]$, $\tau^L \in [0.755, 0.885]$, and $\tau^k \in [0.071, 0.171]$. Appendix D.3 also reports fixed absolute government spending.

get rule returns to households: revenue rises by 3.3 percentage points of GDP, six times the increase at the progressivity optimum. The last column of Table 4 therefore compares the instruments at equal revenue. At the 0.54 percentage points of GDP raised by the progressivity optimum, a higher labor-tax level yields a welfare gain of 1.4% and a higher capital tax 1.2%, against 4.3% for progressivity, three to four times as much. Progressivity is also the only instrument whose best reform does not reduce measured TFP: TFP is 0.004% above its benchmark at the progressivity optimum, 0.48% below it at the best labor-tax-level reform, and essentially unchanged at the capital-tax edge.

Figure 9 shows the underlying policy profiles. Progressivity raises formal labor supply among households in the bottom three quarters of the productivity distribution, while the top quarter changes little until the evasion margin becomes active. A higher labor-tax level lowers the return to formal work throughout the distribution. Formal labor supply falls among lower-productivity households, while households in the top quarter keep all their hours in formal work over the explored range. Informality therefore rises throughout the labor-tax-level experiment. Capital taxation

mainly redistributes from households with high asset income; informality moves by less than a tenth of a percentage point over the explored range.



Notes: Each tax instrument varies separately, with other tax and structural parameters fixed. The labor-tax level $1 - \tau^L$ is the average tax rate at mean formal earnings. A horizontal-axis change of 0.01 is one percentage point. Government consumption remains 15% of output, and the lump-sum transfer balances the budget. Welfare gains are steady-state consumption-equivalent variations (CEV, percent), including leisure, relative to the baseline. Informality (the informal share of hours worked) and the formal-hours share κ are levels; panel (c) splits households by productivity. Panel (d) reports changes in tax revenues relative to GDP. The labor-tax-level profiles are shown up to +9 percentage points; Table 4 reports the interior welfare maximum of this instrument, at +10.

Figure 9: Progressivity compared with labor- and capital-tax levels

The second block of Table 4 separates these comparisons from the use of additional revenue. Here the lump-sum instrument is held fixed, and government consumption absorbs revenue changes. Government consumption has no direct utility value in the model, so the welfare comparison reflects changes in private consumption and leisure. Without further restriction, welfare in this experiment rises as government consumption shrinks, so we require government consumption to remain at least

at its baseline share of output. Higher progressivity still raises welfare by 2.65%, at $\tau^P = 0.115$, where this constraint binds, and reduces informality by 6.0 percentage points. Raising the level of either labor or capital taxation no longer improves welfare: their welfare gains come only from cutting the rate, which the floor on government consumption rules out, so their best admissible reform is the baseline itself. At the increase in revenue/GDP delivered by the progressivity reform they lower welfare by 0.11% and 0.18%, respectively.

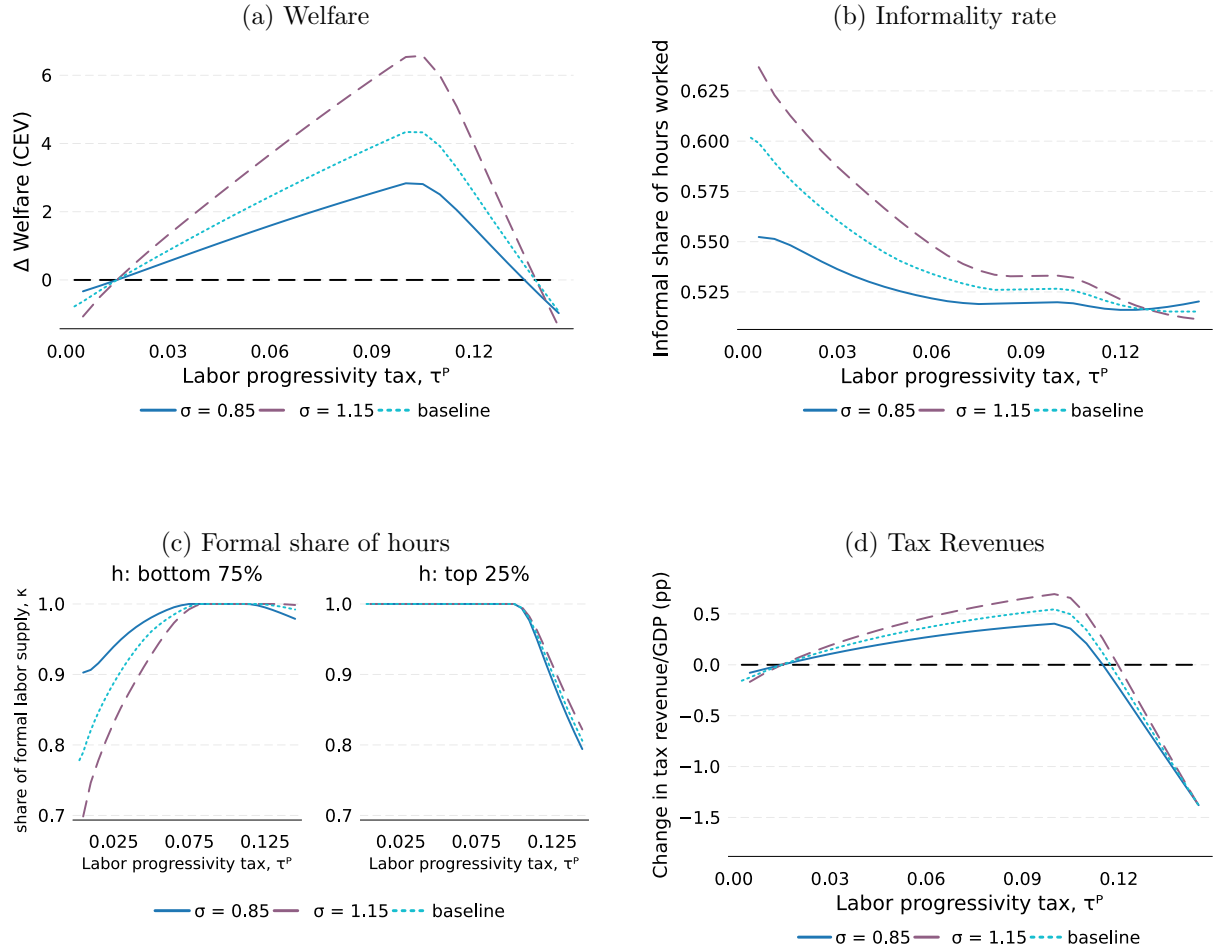
Progressivity consequently offers a distinct combination of redistribution and formalization. Higher labor-tax levels redistribute through their revenue while making formal work less attractive. Capital taxes redistribute with little change in sectoral labor supply. Progressivity redistributes through the formal sector itself: it increases the return to formal work at the bottom, while higher taxes on earnings at the top finance the subsidy. Its welfare gain remains positive even when households receive none of the additional revenue, which is not the case for the level instruments. This comparison identifies the advantage of changing the shape of labor taxation in a high-informality economy.

5.3 Inequality and Optimal Progressivity

We next ask how underlying inequality changes the preferred degree of progressivity and the gains from adopting it. Long-run dispersion in productivity is $\sigma_{LR} = \sigma_h / \sqrt{1 - \rho^2}$. Figure 10 compares a low-inequality economy with $\sigma_{LR} = 0.85$, the baseline with $\sigma_{LR} = 1.00$, and a high-inequality economy with $\sigma_{LR} = 1.15$. All other structural parameters are held fixed, and each reform is evaluated relative to that economy's baseline tax system, with the labor-tax schedule specified relative to that economy's own average formal earnings. The reported gains therefore measure the value of tax reform separately at each level of earnings risk.

These differences generate substantial variation in income and consumption inequality. Moving from the low- to the high-inequality economy raises the labor-income Gini from 0.37 to 0.49 and the variance of log consumption from 0.48 to 0.83. The wealth Gini instead falls from 0.79 to 0.73. Greater productivity risk raises precautionary saving, but equilibrium interest-rate changes and accumulation dynamics can compress the upper tail of wealth (Aiyagari, 1994; Benhabib et al., 2011; Huggett, 1996). The experiment therefore concerns greater earnings risk and consumption inequality, rather than a uniform increase in every measure of dispersion.

Welfare remains hump-shaped in progressivity in all three economies. The optimum barely moves: $\tau^P = 0.10$, 0.10, and 0.105 in the low-, baseline-, and high-inequality economies. The welfare



Notes: The economies differ in the long-run standard deviation of log productivity, $\sigma_{LR} = 0.85, 1.00$ (baseline), and 1.15; other structural parameters are fixed. Each economy's labor-tax schedule is specified relative to its own average gross formal earnings at baseline progressivity, held fixed as τ^P varies. Government consumption remains 15% of output, and the lump-sum transfer balances the budget. Welfare gains are steady-state consumption-equivalent variations (CEV, percent), including leisure, relative to each economy's own equilibrium at baseline progressivity $\tau^P = 0.015$. Informality (the informal share of hours worked) and the formal-hours share κ are levels; panel (c) splits households by productivity. Revenue changes are relative to each economy's own baseline.

Figure 10: Inequality and the welfare stakes of progressivity

stakes change much more. The gain from the optimal schedule rises from 2.8% of consumption to 4.3% and 6.6%, respectively. Welfare rises faster at low progressivity and falls more sharply beyond the peak when inequality is higher.

Both margins contribute to this steeper profile. Greater productivity dispersion puts more households in the low-productivity tail, where formal jobs are scarce and informal returns are relatively high. These households have more to gain from subsidies to formal work. At the top, a more dispersed distribution increases the importance of households whose formal labor supply responds to high marginal rates. Progressivity therefore generates larger reallocations at both ends of the distribution. Panels 10c and 10d show the associated formal labor supply and revenue profiles. Greater dispersion increases the concentration of taxable income at the top. As high-productivity households respond to marginal rates, revenue also becomes more sensitive to progressivity, reinforcing the steeper welfare profile.

The stronger insurance motive is thus met by a stronger evasion response. In our experiments, the two effects largely offset in the location of the optimum, while increasing the gain from adopting it. The changes in labor-income inequality at the optimum are comparatively similar: labor-income Ginis fall by 1.5 to 1.9 percentage points across the three economies, while the variance of log consumption falls by 0.03 to 0.07, more in the more unequal economies. Higher inequality raises the welfare stakes of tax design much more than the preferred degree of progressivity. The standard prescription is that higher uninsurable earnings risk calls for a more progressive schedule (Heathcote et al., 2017, 2020); here the optimum rises only weakly with risk.

5.4 Robustness of the Policy Results

The welfare-maximizing degree of progressivity is similar across the three government budget rules in Appendix D.2. It is 0.10 under both the benchmark rule and fixed absolute spending, and 0.115 when the transfer is fixed and government consumption must remain at least at its baseline share of output. Informality falls by 5.4 to 6.0 percentage points in each case. Welfare gains are 4.34%, 3.04%, and 2.65%, respectively. With G/Y fixed, changes in output also change the resources absorbed by government consumption; with G fixed, this quantity does not adjust. Without the floor on G/Y , welfare under the fixed transfer keeps rising with progressivity as government consumption shrinks. The preferred schedule is therefore stable across these fiscal adjustments, while the gains depend on how the government uses revenue and how spending changes with output.

Alternative calibrations in Appendix C assess the role of earnings risk and informal returns.

Table 5 compares their preferred tax schedules and welfare gains under the benchmark and fixed-spending rules. Adding the estimated sector-specific transitory component yields an optimum of 0.06, against that calibration’s benchmark of 0.018. The welfare gain falls to 1.85% under the benchmark budget rule. Fixing the curvature of informal returns at its panel estimate, together with the panel estimate of earnings persistence, yields an optimum of 0.125, against a benchmark of 0.016, with a welfare gain of 1.2%. Under fixed absolute spending, that calibration’s optimum is 0.09. In both alternatives, optimal progressivity remains several times above its benchmark.

Table 5: Robustness of optimal progressivity

Calibration	Benchmark	G/Y fixed		G fixed	
	τ^P	Optimal τ^P	CEV (%)	Optimal τ^P	CEV (%)
Baseline	0.015	0.100	4.34	0.100	3.04
Fixed ξ	0.016	0.125	1.24	0.090	0.43
Transitory earnings risk	0.018	0.060	1.85	0.060	1.26

Notes: Each calibration is re-fitted before varying labor-tax progressivity τ^P . Welfare gains are steady-state consumption-equivalent variations (CEV), including leisure, relative to that calibration’s benchmark. Lump-sum transfers balance the budget with government consumption fixed relative to output (G/Y) or in levels (G). The fixed- ξ calibration sets informal-return curvature at its panel estimate, $\xi = 0.80$, and annual earnings persistence at $\rho = 0.944$. The transitory-risk calibration matches sector-specific transitory earnings variances from Mexico’s ENOE panel using a common shock, keeps the baseline persistence, and re-fits the dispersion of the persistent component. Appendix C reports calibration details.

The alternatives also reveal which features affect the size of the gains. Households can draw down assets after a temporary earnings loss. A persistent loss reduces resources over many periods and is harder to offset through saving alone, giving taxes a larger role in insuring consumption. Matching the panel risk estimates therefore changes the wealth and consumption distributions as well as sectoral labor allocation. The transitory-risk calibration assumes a common shock with different sectoral loadings; the panel identifies the sectoral variances but not their correlation. Under the benchmark fiscal rule and the common-shock assumption, both alternative calibrations favor substantially higher progressivity. The welfare gains and sectoral allocation responses depend more strongly on the calibration. Informality falls by 7.2 percentage points at the optimum with ξ fixed, but rises by 0.3 points with transitory risk, where almost every worker with a formal job already works only formal hours at the benchmark, thus muting the inclusion margin, so progressivity has no informal hours to draw into formal work (Appendix C.1).

Finally, Appendix D.5 varies the wage rule, labor-supply elasticity, and informal returns one at a time. Changing the wage-rule weight by ± 0.15 moves optimal progressivity by 0.005 to 0.01. The

labor-supply elasticity matters more for the size of the gains: halving the Frisch elasticity raises the optimum to 0.12 and halves the welfare gain, while doubling it lowers the optimum to 0.09 and raises the gain to 5.0%. Informal returns move the optimum most: lowering ξ from 0.92 to 0.87 raises it to about 0.135, while raising ξ to 0.97 lowers it to 0.07 and keeps about 70% of the baseline gain. When informal earnings rise more strongly with ability, high-productivity households can avoid taxes at lower cost. Informal returns therefore matter for tax design through their effect on evasion.

6 Conclusion

Progressive labor taxation can improve welfare and reduce informality. Negative income taxes draw low earners into the formal tax base, while higher marginal rates at the top eventually shift labor toward untaxed informal work. In our model calibrated to Mexico, the welfare-maximizing degree of progressivity is nearly an order of magnitude above the current level. The reform raises steady-state welfare by 4.3% of consumption and lowers the informal share of hours worked by 5.4 percentage points.

The comparison across fiscal instruments shows why the shape of taxation matters. Higher labor-tax levels redistribute through their rebated revenue but increase informality. Higher capital taxes redistribute with little change in sectoral labor allocation. Progressivity combines redistribution with a substantial increase in formality, which no other instrument delivers. At equal revenue it yields three to four times the welfare gain of either level instrument, and its welfare gains persist when additional revenue is not returned to households. Higher inequality increases the gains from choosing the right schedule, while changing the preferred degree of progressivity relatively little.

These findings concern the household labor-supply margin. Firms' formalization decisions and income pooling across household members provide additional channels through which taxes can affect informality. Adding the transitory earnings risk measured in the panel also reveals a tension with the persistent risk needed to match wealth and consumption inequality. Alternative calibrations preserve the scope for a substantial increase in progressivity, while changing the size of its gains. The results show that a large informal sector leaves substantial scope for redistribution when tax design raises the return to formal work at the bottom.

References

- Abel, M., Carranza, E., Geronimo, K., and Ortega, M. E. (2022). Can temporary wage incentives increase formal employment? experimental evidence from mexico. IZA Discussion Paper 15740, IZA Institute of Labor Economics.
- Aiyagari, S. R. (1994). Uninsured idiosyncratic risk and aggregate saving. *The Quarterly Journal of Economics*, 109(3):659–684.
- Albrecht, J., Navarro, L., and Vroman, S. (2009). The effects of labour market policies in an economy with an informal sector. *The Economic Journal*, 119(539):1105–1129.
- Bachas, P., Brockmeyer, A., Dom, R., and Semelet, C. (2023). Effective tax rates and firm size. Policy Research Working Paper 10312, World Bank, Washington, DC.
- Bachas, P., Gadenne, L., and Jensen, A. (2024). Informality, consumption taxes, and redistribution. *The Review of Economic Studies*, 91(5):2604–2634.
- Banerjee, A., Hanna, R., Olken, B. A., and Lisker, D. S. (2023). Social protection in the developing world.
- Benabou, R. (2002). Tax and education policy in a heterogeneous-agent economy: What levels of redistribution maximize growth and efficiency? *Econometrica*, 70(2):481–517.
- Benhabib, J., Bisin, A., and Zhu, S. (2011). The distribution of wealth and fiscal policy in economies with finitely lived agents. *Econometrica*, 79(1):123–157.
- Bérgolo, M. and Cruces, G. (2021). The anatomy of behavioral responses to social assistance when informal employment is high. *Journal of Public Economics*, 193:104313.
- Bobba, M., Flabbi, L., Levy, S., and Tejada, M. (2021). Labor market search, informality, and on-the-job human capital accumulation. *Journal of Econometrics*, 223(2):433–453.
- Bosch, M. and Campos-Vázquez, R. M. (2014). The trade-offs of welfare policies in labor markets with informal jobs: The case of the Seguro Popular program in mexico. *American Economic Journal: Economic Policy*, 6(4):71–99.
- Bosch, M. and Esteban-Pretel, J. (2012). Job creation and job destruction in the presence of informal markets. *Journal of Development Economics*, 98(2):270–286.

- Castaneda, A., Diaz-Gimenez, J., and Rios-Rull, J.-V. (2003). Accounting for the us earnings and wealth inequality. *Journal of political economy*, 111(4):818–857.
- Chong, A. and Gradstein, M. (2007). Inequality and informality. *Journal of Public Economics*, 91.
- Conesa, J. C. and Krueger, D. (2006). On the optimal progressivity of the income tax code. *Journal of Monetary Economics*, 53(7):1425–1450.
- Den Haan, W. J., Rendahl, P., and Riegler, M. (2018). Unemployment (fears) and deflationary spirals. *Journal of the European Economic Association*, 16(5):1281–1349.
- Departamento Nacional de Estadística (2009). *Metodología informalidad Gran Encuesta Integrada de hogares -- GEIH*. Departamento Nacional de Estadística.
- Doligalski, P. and Rojas, L. E. (2023). Optimal redistribution with a shadow economy. *Theoretical Economics*, 18:749–791.
- Donovan, K., Lu, W. J., and Schoellman, T. (2023). Labor market dynamics and development. *The Quarterly Journal of Economics*, 138(4):2287–2325.
- Elgin, C. and Oztunali, O. (2012). Shadow Economies around the World: Model Based Estimates. Working Papers 2012/05, Bogazici University, Department of Economics.
- Engbom, N., Gonzaga, G., Moser, C., and Olivieri, R. (2022). Earnings inequality and dynamics in the presence of informality: The case of brazil. *Quantitative Economics*, 13(4):1405–1446.
- Engbom, N., Malmberg, H., Porzio, T., Rossi, F., and Schoellman, T. (2025). Economic development according to Chandler. NBER Working Paper 34483, National Bureau of Economic Research.
- Ferriere, A., Grübener, P., Navarro, G., and Vardishvili, O. (2023). On the optimal design of transfers and income tax progressivity. *Journal of Political Economy Macroeconomics*, 1:276–333.
- Galiani, S. and Weinschelbaum, F. (2012). Modeling informality formally: Households and firms. *Economic Inquiry*, 50.
- Gerard, F. and Gonzaga, G. (2021). Informal labor and the efficiency cost of social programs: Evidence from unemployment insurance in Brazil. *American Economic Journal: Economic Policy*, 13(3):167–206.

- Gomes, D. B. P., Iachan, F. S., and Santos, C. (2020). Labor earnings dynamics in a developing economy with a large informal sector. *Journal of Economic Dynamics and Control*, 113:103854.
- Gornemann, N., Kuester, K., and Nakajima, M. (2016). Doves for the rich, hawks for the poor? distributional consequences of monetary policy. CEPR Discussion Papers DP11233, Centre for Economic Policy Research.
- Gorodnichenko, Y., Martinez-Vazquez, J., and Sabirianova Peter, K. (2009). Myth and reality of flat tax reform: Micro estimates of tax evasion response and welfare effects in russia. *Journal of Political Economy*, 117(3):504–554.
- Graves, S. (2025). Does unemployment risk affect business cycle dynamics? *American Economic Journal: Macroeconomics*, 17(2):65–100.
- Gunter, S. (2013). State earned income tax credits and participation in regular and informal work. *National Tax Journal*, 66(1):33–62.
- Heathcote, J., Storesletten, K., and Violante, G. L. (2014). Consumption and labor supply with partial insurance. *American Economic Review*, 104(7):2075–2126.
- Heathcote, J., Storesletten, K., and Violante, G. L. (2017). Optimal tax progressivity: An analytical framework. *The Quarterly Journal of Economics*, 132(4):1693–1754.
- Heathcote, J., Storesletten, K., and Violante, G. L. (2020). Presidential address 2019: How should tax progressivity respond to rising income inequality? *Journal of the European Economic Association*, 18(6):2715–2754.
- Holter, H. A., Krueger, D., and Stepanchuk, S. (2019). How do tax progressivity and household heterogeneity affect laffer curves? *Quantitative Economics*, 10(4):1317–1356.
- Huggett, M. (1996). Wealth distribution in life-cycle economies. *Journal of Monetary Economics*, 38(3):469–494.
- Instituto Brasileiro de Geografia e Estatística (2020). *Síntese de indicadores sociais uma análise das condições de vida da população brasileira 2020*. Instituto Brasileiro de Geografia e Estatística.
- Instituto Nacional de Estadística, Geografía e Informática (2014). *La informalidad laboral. Encuesta Nacional de Ocupación y Empleo. Marco conceptual y metodológico*. Instituto Nacional de Estadística, Geografía e Informática.

- International Labor Organization (2003). *Seventeenth International Conference of Labour Statisticians (Geneva, 24 November - 3 December 2003), Report of the Conference; Doc. ICLS/17/2003/R*, chapter Guidelines concerning a statistical definition of informal employment, endorsed by the Seventeenth International Conference of Labour Statisticians (November-December 2003). International Labor Organization.
- Kindermann, F. and Krueger, D. (2022). High marginal tax rates on the top 1 percent? lessons from a life-cycle model with idiosyncratic income risk. *American Economic Journal: Macroeconomics*, 14(1).
- King, R. G., Plosser, C. I., and Rebelo, S. T. (1988). Production, growth and business cycles: I. the basic neoclassical model. *Journal of monetary Economics*, 21(2-3):195–232.
- Kleven, H. J., Kreiner, C. T., and Saez, E. (2016). Why can modern governments tax so much? An agency model of firms as fiscal intermediaries. *Economica*, 83(330):219–246.
- Kose, M. A., Elgin, C., Ohnsorge, F., and Yu, S. (2021). Understanding Informality. CEPR Discussion Papers 16497, C.E.P.R. Discussion Papers.
- La Porta, R. and Shleifer, A. (2014). Informality and Development. *Journal of Economic Perspectives*, 28(3):109–126.
- Levy, S. (2008). *Good Intentions, Bad Outcomes*. Brookings Institutions Press.
- Leyva, G. and Urrutia, C. (2020). Informality, labor regulation, and the business cycle. *Journal of International Economics*, 126(C).
- Maloney, W. F. (2004). Informality revisited. *World Development*, 32.
- Meghir, C., Narita, R., and Robin, J.-M. (2015). Wages and informality in developing countries. *American Economic Review*, 105(4):1509–1546.
- OECD (2025). *Revenue Statistics 2025: Disentangling Personal Income Tax Revenue in OECD Countries*. OECD Publishing, Paris.
- OECD (2026). *Taxing Wages 2026: The Progressivity of Labour Taxation in OECD Countries*. OECD Publishing, Paris.
- Perry, G. E., Maloney, W. F., Arias, O. S., Fajnzylber, P., Mason, A. D., and Saavedra-Chanduvi, J. (2007). *Informality: Exit and Exclusion*. World Bank Publications.

- Rosser, J., Rosser, M. V., and Ahmed, E. (2000). Income inequality and the informal economy in transition economies. *Journal of Comparative Economics*, 28(1):156–171.
- Schneider, F., Buehn, A., and Montenegro, C. E. (2010). New estimates for the shadow economies all over the world. *International Economic Journal*, 24(4):443–461.
- Schoar, A. (2009). The divide between subsistence and transformational entrepreneurship. *Innovation Policy and the Economy*, 10:57–81.
- Slonimczyk, F. (2012). The effect of taxation on informal employment: Evidence from the russian flat tax reform. *Research in Labor Economics*, 34:55–99.
- UBS (2025). Global wealth report 2025. Technical report, UBS.
- Ulyssea, G. (2018). Firms, informality, and development: Theory and evidence from Brazil. *American Economic Review*, 108.
- Ulyssea, G. (2020). Informality: Causes and consequences for development. *Annual Review of Economics*, 12.

A Data Sources

In this appendix, we discuss the sources of information to construct labor market indicators and inequality, and the details entailed in each general definition. Overall, our main sources of information are household surveys for Brazil, Colombia, Mexico and Peru. In what follows, we describe some considerations for the construction of the statistics used throughout the paper. In all countries, we restrict our samples to people age 15 and older.

A.1 Brazil

The main source of data for the analysis of labor market outcomes over the income distribution is the *Pesquisa Nacional por Amostra de Domicílios Continua* (PNADC) carried out by the Brazilian Statistics Bureau (IBGE). The PNADC is a nationally representative survey of Brazilian households that provides information of the characteristics of the households and their members, and their labor market status. All households in the PNADC are interviewed by five consecutive quarters. The current version of the PNADC has run continuously since 2012, and the data are publicly available on the IBGE website.

Throughout the paper (see Section 2), we use information of labor force participation, employment, and unemployment, formality status (formal or informal) and labor earnings. Labor force status follows standard classification used by the IBGE and is available for all people age 14 and older.¹⁴ For those employed, the survey asks questions to characterize the job. We proxy formality status based on a definition used by the IBGE (IBGE, 2020). Using the worker’s main job, a worker is considered a formal-sector worker if he or she works for a person or a firm and reports having a registered job, which in Brazil is mandatory and implies having a work card (*carteira de trabalho*) signed by the workers’ employer. For self-employed workers and employers, we define the formality status based on whether they have their businesses registered with the tax authorities (CNPJ).

The PNADC also collects information about the workers’ labor earnings. Earnings are self-reported, cover all jobs (including salaried and independent work), and refer to monthly earnings. Earnings are in current Brazilian Reais (BRL\$).

¹⁴The definitions among countries are very similar and resemble those used by the US Current Population Survey.

A.2 Colombia

The source of data for Colombian labor market is the *Gran Encuesta Integrada de Hogares* (GEIH) carried out by the Colombian Statistics Bureau (DANE). The GEIH is a national representative survey that collects information of labor market outcomes and social conditions of the households. It is the official source of employment statistics, incidence of monetary poverty and income inequality. The GEIH is a repeated cross section dataset collected monthly and has run continuously since 2008. Data are publicly available in the DANE website.

We define the labor force status following DANE’s standard definitions, which are available for all people age 12 and older (10 and older in rural areas). We use DANE’s official definition for informal employment, which defines an informal worker as a person who either works in small production units (up to five workers), excluding public-sector workers and professional self-employed (DANE, 2009). This definition applies for all workers. Although the official statistics is reported for urban areas only, we also apply the definition of informality for workers in rural areas.

Regarding labor earnings, we use monthly labor earnings from the main job, reported in Colombian Pesos (COP\$).

In contrast to the other countries, the GEIH is a repeated cross section dataset, which implies that is not possible to follow individuals over time to get labor market transitions by comparing individuals over time. Nonetheless, the GEIH includes retrospective questions (e.g., tenure on the job, the type of job the person had before, how many weeks an unemployed person has been looking for a job and so on) that allow to infer labor transitions.

A.3 Mexico

The main source of data for Mexico is the *Encuesta Nacional de Ocupación y Empleo* (ENOE) carried out by the Mexican Statistics Bureau (INEGI). The ENOE is a nationally representative survey of Mexican households used to characterize their labor market outcomes. The survey is the official source of labor force statistics. In contrast to other datasets used in this analysis, the ENOE is not the dataset used to measure monetary poverty and income inequality. As in the case of Brazil’s PNADC, the ENOE interviews all households by five consecutive quarters. The ENOE has run continuously since 2005. Data are publicly available on the INEGI website.

In the information of labor market status (employed, unemployed, out of the labor force) we follow the classification elaborated by INEGI. The data about labor market outcomes are available for all people age 12 and older, however, official statistics are reported for people age 15 and more

only. Regarding the informality definition, the ENOE contains a variable to classify workers between formal and informal employment. As in the case of Colombia, the definition in Mexico follows the guidelines proposed by the International Labor Organization for the measurement of informal employment (ILO, 2003). Broadly speaking, an informal worker is a person working in a job outside the scope of government regulation – and therefore is unprotected by labor legislation – either because he or she works in a small, low-productivity unit (typically unincorporated, including self-employment and subsistence activities), his or her job are uncovered by the law, or their employer keep his or her job off-the-books (INEGI, 2014). Compared to Colombia and Brazil, the operational definition of informality in Mexico is more detailed, as it includes specific questions to characterize the features of an informal job.

We use monthly labor earnings from the main job, reported in Mexican Pesos (MXN\$).

In addition to the cross sections, the results in Section 2.3 and the calibrations of Appendix C exploit the ENOE’s rotating-panel design, linking households and their members across the sixteen quarters from 2021Q1 to 2024Q4, a window chosen to fall entirely within the survey’s current redesigned version. Individuals are linked with the standard household-person key augmented by the dwelling and sampling-unit identifiers, which removes the spurious matches the standard key alone produces; in the resulting file, duplicate matches within a quarter are below 0.2 percent and the sex of linked individuals is consistent in essentially all pairs. The linked panel contains about 5.1 million person-quarter observations on 1.6 million individuals. The persistence estimates use log hourly earnings from the main job for formal salaried workers aged 25 to 60, roughly 677,000 wage observations on 318,000 workers, and fit a persistent AR(1) component plus serially uncorrelated noise to the autocovariances at lags zero to four. All panel statistics use the survey’s sampling weights, and the construction is documented in the replication package.

Informal returns to ability: the switcher regression

Equation (1) relates the formal and informal hourly wages of a worker with given ability. We estimate it on workers aged 25 to 60 whom the linked panel observes in informal work at least once and in formal work at least twice within their five interviews. Wages are log hourly earnings from the main job, trimmed at the 0.5th and 99.5th percentiles and residualized on a quadratic in age, sex, a quadratic in years of schooling, and state, locality-size, and quarter dummies.

Two problems rule out simpler estimators. Comparing the dispersion of earnings across sectors uses different people in each sector, and is contaminated by exactly the selection on ability

that the model is about; it returns a value above one. Regressing a switcher’s informal wage on their formal wage by OLS compares the same worker across sectors, but is attenuated by measurement error in survey earnings and returns 0.36. We therefore estimate (1) instrumenting the worker’s formal log wage with a *second* formal observation of the same worker at a different date, thus $\hat{\xi} = \text{cov}(\ln W_{it}^s, \ln W_{ir}^f) / \text{cov}(\ln W_{is}^f, \ln W_{ir}^f)$ with $r \neq s$, which removes attenuation from serially uncorrelated measurement error under the assumption that the instrument is uncorrelated with intervening ability innovations in the switcher sample. Because ability mean-reverts between interviews, the two covariances are comparable only when the informal and the formal observation are equally distant from the instrument; when they are not, we rescale by the estimated quarterly persistence of residual wages, 0.970, and we also report the lag-matched subsample in which no rescaling is needed. Standard errors are from a bootstrap clustered at the worker level, and all estimates use the survey’s sampling weights.

The estimand is the elasticity for workers observed changing sectors within the panel. Selection on the level of ability changes the distribution of the regressor, while selection on ability innovations at the time of a sector change can affect the validity of the instrument. The interpretation of the estimate as the structural curvature therefore requires the stated orthogonality condition within the switcher sample.

Table A.1 reports the estimates. Across wage concepts, informality definitions, and the treatment of mean reversion, the instrumented elasticity lies between 0.75 and 0.87, and every confidence interval excludes one. The lag-matched estimates agree with the rescaled ones, so the results are robust to the mean-reversion-adjustment assumption. The last two rows also identify the intercept. The last row, which needs no mean-reversion adjustment, is the specification that calibration (b) of Appendix C imposes. The raw intercept depends on the units in which wages are measured, so we report it in unit-free form, as the predicted informal-to-formal wage ratio at the mean formal wage: in the first column, 0.940 (s.e. 0.005) in the penultimate row and 0.921 (s.e. 0.007) in the last row, evaluated at the mean over all formal workers, and 0.949 (s.e. 0.005) and 0.930 (s.e. 0.007) at the mean over switchers. The social-security definition gives 0.949 and 0.922 at the mean over all formal workers; monthly earnings give lower ratios, 0.865 and 0.871 (0.868 and 0.869 under the social-security definition), because informal main jobs have fewer hours. The scale parameter Ξ is not separately identified, since the intercept also contains the equilibrium prices w^s and w^f , which the survey does not observe; the model computes the same wage ratio at its own prices, and calibration (b) of Appendix C targets the last row’s ratio at the mean over all formal workers, 0.921.

Table A.1: Estimates of the curvature of informal returns, ξ

Informality definition:	Hourly wage		Monthly earnings	
	Baseline	Soc. security	Baseline	Soc. security
Cross-sectional variance ratio	1.048	1.048	1.232	1.232
Within-person OLS	0.355	0.355	0.413	0.411
Within-person IV	0.840 (0.027)	0.832 (0.027)	0.852 (0.027)	0.843 (0.026)
Within-person IV, lag-matched	0.827 (0.036)	0.818 (0.030)	0.866 (0.036)	0.859 (0.031)
Joint IV, with informal self-employment	0.784 (0.022)	0.747 (0.024)	0.789 (0.021)	0.785 (0.023)
Joint IV, salaried, lag-matched	0.813 (0.037)	0.805 (0.036)	0.865 (0.034)	0.858 (0.032)

Notes: ENOE linked panel, 2021Q1–2024Q4, workers aged 25–60. Regressions relate informal to formal log earnings. The salaried regressions use earnings residualized within sector: 18,990 switchers contribute 102,102 wage triples, including 14,960 lag-matched triples. IV uses a second formal observation of the same worker; lag matching places both wage observations equally far from the instrument. “Baseline” uses the ENOE formality classification; “Soc. security” classifies jobs without social-security coverage as informal, so almost all self-employment is informal under it (99.8% of own-account workers and 99.4% of employers, against 87% and 55% under the baseline). The penultimate row adds informal self-employment (31,048 switchers, 167,092 triples; 23,766 and 130,444 under the social-security definition), uses common covariate coefficients, and jointly estimates the slope and intercept; the last row applies the same joint estimator to salaried workers and lag-matched triples (11,992 switchers, 15,011 triples; 11,894 and 14,886). Sample sizes are the same for hourly wages and monthly earnings. Only the “Within-person IV” row adjusts for mean reversion. All estimates use survey weights; parentheses report worker-clustered bootstrap standard errors.

Because survey earnings are net of taxes, that calibration scales the last row’s slope by $1 - \tau^P$ to obtain the curvature with respect to gross earnings, $\xi = 0.80$.

An independent pattern in the same data corroborates $\xi < 1$ without relying on the instrument. Relative to formal workers who start from the same wage decile and stay formal, the one-year wage loss of those who move into informality is between 0.07 and 0.10 log points from the second to the sixth decile and then grows steadily to 0.35 log points in the top decile. If informal earnings were proportional to ability, the loss would not depend on where in the distribution the worker starts; with $\xi < 1$ it should widen by $(1 - \xi)$ times the ability gap between deciles. Formal wages differ by 1.8 log points between the second and the top decile, about three fifths of which is persistent, so ξ between 0.78 and 0.84 predicts a widening of roughly 0.2 log points; the observed widening is 0.26.

A.4 Peru

The main source of data for Peru is the Encuesta Nacional de Hogares (ENAHOG) carried out by the Peruvian Statistics Bureau (INEI). The ENAHOG is a nationally representative survey of Peruvian households used to characterize labor market outcomes, and households’ living standards. It is the official source of employment statistics, incidence of monetary poverty and household income inequality. The ENAHOG has a mixed structure in which a subsample of households (about 25% of total) are interviewed every year for up to five years. ENAHOG data are available continuously since 2008. Data are publicly available in the INEI website.

Information about labor market outcomes is available for all people age 14 and older, and we follow the classification made by the INEI. As in the case of the ENOE in Mexico, the ENAHOG includes detailed information that allow to classify an informal worker depending on the type of production unit they work and whether they are covered by labor regulation.

The ENAHOG collects information about the workers’ labor earnings in all jobs to compute measures of annual labor earnings. Throughout the paper, we take this measure of annual labor earnings divided by 12 to get a measure of monthly labor earnings. Earnings are in current Peruvian Soles (PEN\$).

B Inequality and Informality

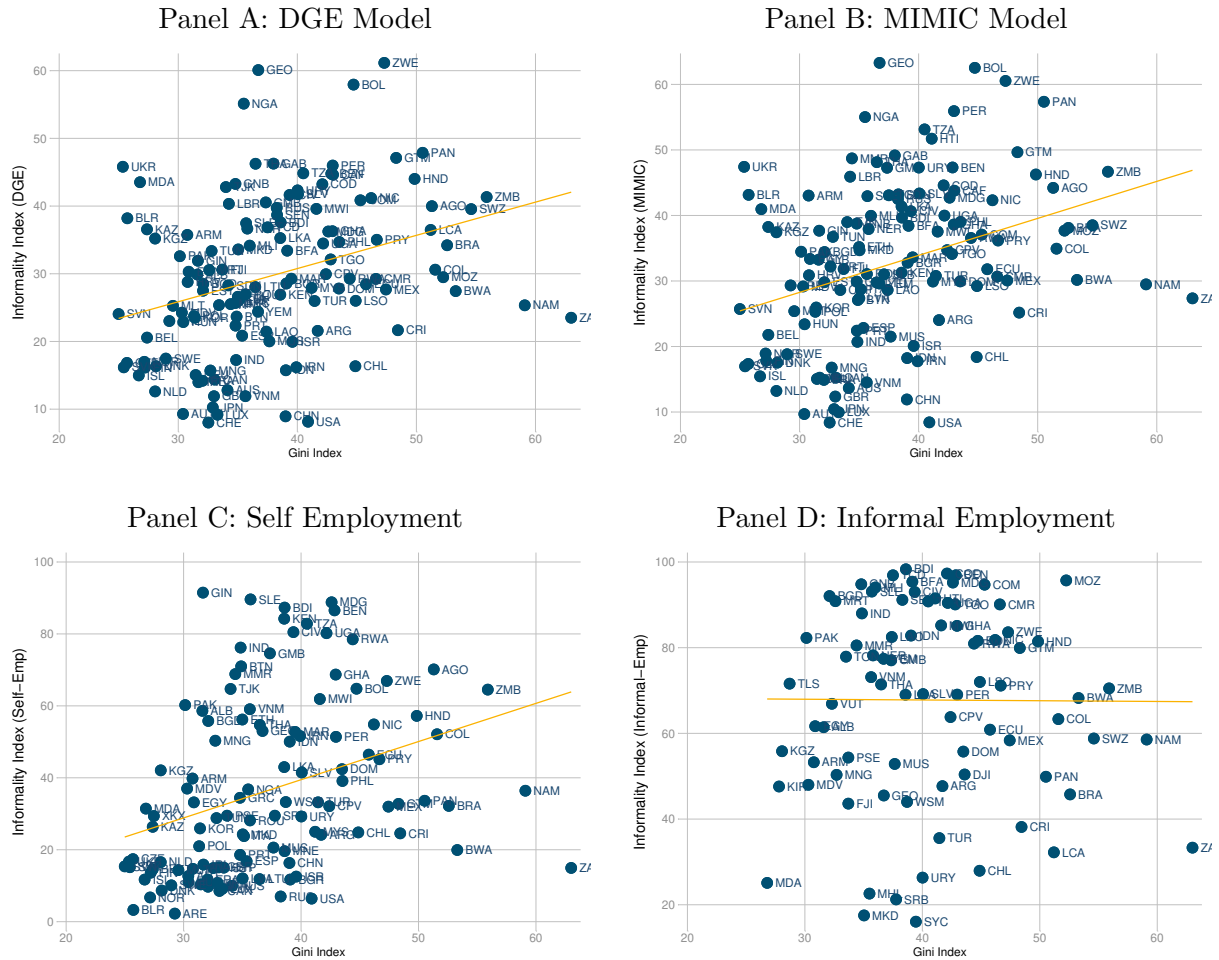
Several confounding variables, such as income levels or human capital in a country, might influence the relationship between inequality and informality. A thorough analysis of this relationship is beyond the scope of this paper; here, we simply compare the relationship between the Gini index of income inequality reported by the World Bank with informality measures produced and updated by [Kose et al. \(2021\)](#). Given its multidimensional nature, we employ multiple direct and indirect estimation methods to account for the difficulty in measuring informality. We focus on measures with sufficient variation across countries and time.

The analysis includes two indirect measures: The general equilibrium model (*DGE*) and the multiple indicator multiple causes model (*MIMIC*). The *DGE* model estimates the size of the informal economy as a percentage of GDP using the approach in [Elgin and Oztunali \(2012\)](#), which is available for 158 countries. The *MIMIC* model measures the size of the informal sector as a percentage of GDP using the model of [Schneider et al. \(2010\)](#), available for 160 countries.

We also include two direct measures of informality based on country surveys. The first measure is the share of self-employment in total employment (*Self-Emp*). This category encompasses four sub-categories of jobs: employers, own-account workers, members of producers cooperatives, and contributing family workers. The share of *Self-Emp* is obtained from the labor force and household surveys for a sample of 180 countries. The second direct measure corresponds to the share of informal employment in percent of total employment (*Informal-Emp*), obtained from household and labor surveys and available for 72 countries.

Figure [B.1](#) presents the cross-sectional scatter plot between the Gini index average and the four informality measures from 2002 to 2022. We observe a positive correlation between inequality and the indirect measures of informality and the share of self-employment in Panels A to C. There is little correlation between inequality and the measure of informal employment in Panel D. The number of observations for this measure is also smaller.

Table [B.1](#) reports panel regressions with and without country fixed effects. Both indirect measures remain positively associated with inequality. With country effects, the self-employment coefficient becomes small and statistically insignificant, while the informal-employment coefficient becomes statistically significant.



Notes: The figure compares countries using data from 2002–2022. The Gini index is on a 0–100 scale. DGE and MIMIC measure informal output as a percentage of GDP; Self-Emp and Informal-Emp measure self-employment and informal employment as percentages of total employment. Sources: World Development Indicators (World Bank) and Kose et al. (2021).

Figure B.1: Cross-sectional correlations between inequality and informality

Table B.1: Panel Regressions of Informality on Inequality, 1990–2022

	Informality Measures							
	Indirect				Direct			
	<i>DGE</i>		<i>MIMIC</i>		<i>Self-Emp</i>		<i>Informal-Emp</i>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Gini Index	0.636*** (0.13)	0.222*** (0.05)	0.688*** (0.14)	0.215*** (0.03)	1.017*** (0.14)	0.105 (0.12)	0.415 (0.33)	0.939** (0.42)
Observations	1746	1746	1673	1673	1530	1530	329	329
Country FE	No	Yes	No	Yes	No	Yes	No	Yes

Notes: Each column reports the coefficient on the Gini index in a regression of the indicated informality measure, with country fixed effects where shown. DGE and MIMIC measure informal output relative to GDP; Self-Emp and Informal-Emp measure self-employment and informal employment relative to total employment.
Source: World Development Indicators (World Bank) and [Kose et al. \(2021\)](#).

C Robustness Analysis

This appendix asks whether the welfare-maximizing progressivity of Section 5, about six-and-a-half times the calibrated benchmark, depends on two choices of the baseline calibration that the targeted moments do not settle: the income process, which the baseline parameterizes rather than estimates (Section 4), and the curvature of informal returns to ability ξ , which the calibration fits at 0.92 while our panel estimates lie between 0.75 and 0.87 (Appendix A.3). We re-calibrate the model replacing each choice with its ENOE-panel counterpart, one block at a time: alternative calibration (a) adds the sector-specific transitory earnings risk measured in the panel, alternative calibration (b) fixes ξ at the panel estimate and imposes the panel’s earnings persistence. Each economy is re-fitted to the baseline’s targeted moments, and each subsection compares the fit and policy results with the baseline. In both, the ratio of the welfare-maximizing progressivity to the benchmark is of the same order as in the baseline: about three in calibration (a) and about eight in calibration (b).

C.1 Alternative calibration (a): sector-specific transitory earnings risk

Using the ENOE panel, we estimate each sector’s cross-sectional variance of log earnings and decompose it into persistent and transitory components: of the formal variance of 0.325, 0.132 is transitory; of the informal 0.674, 0.313 is. The transitory component is 2.4 times larger in the informal sector, and roughly half the sectoral variance gap is transitory rather than persistent: of the gap of 0.349, 0.180 is transitory, and the baseline targets only its persistent component, 0.169. Alternative calibration (a) adds the measured sectoral variances to the baseline through an i.i.d. multiplicative earnings shock driven by a single common factor with sector-specific loadings, so that $\text{Var}(\ln \varepsilon)$ matches the panel value in each sector and the transitory component of informal earnings is more volatile, with the ergodic dispersion of the persistent component σ_{LR} left free for the moment set to determine.¹⁵ Every other ingredient is the baseline’s, and the targeted moments are those of Section 4.7, except that the variance-gap target is the total gap, 0.349, rather than its persistent component, since the model’s sectoral earnings now include the transitory component.

With the transitory layer imposed, the calibration chooses $\sigma_{LR} = 0.785$, below the baseline’s 1.00, matches the total sectoral variance gap (0.349 against 0.349), and places the curvature of informal returns at an interior $\xi = 0.98$. It also lowers the productivity of informal relative to

¹⁵The panel identifies the two sectoral transitory variances but not their correlation, which must be chosen. We use a single common factor with sector-specific loadings (correlation one). With uncorrelated draws across sectors, hedging the draw becomes the first-order motive of the allocation of labor across sectors, dwarfing the inclusion and evasion margins through which progressivity operates, and the welfare profile over progressivity is flat.

formal work ($\kappa/\Omega = 0.62$, against 0.88 in the baseline) and raises the vacancy cost about sevenfold, so that the labor agency’s surplus on a formal match is much larger relative to the informal wage than in the baseline (0.44 against 0.05). Three other aspects of the fit deteriorate. First, wealth and consumption inequality and the capital-output ratio fall below their data targets (Gini_w 0.66 against 0.72, Gini_c 0.37 against 0.41, K/Y 2.31 against 2.44), because transitory shocks are self-insured through the asset buffer and are less effective than persistent ones in generating precautionary savings. Second, informality is below the data, with an informal headcount of 0.49 against 0.54 (an informal share of hours of 0.50 against 0.51), while the job-finding rate is above it (0.18 against 0.16) and the mean log-earnings gap falls to 0.64 against 0.72, and the informal-to-formal net wage ratio falls to 0.72 against 0.92. The added transitory component improves the fit of earnings dispersion but makes the two labor-market targets harder to match jointly. Third, no worker splits hours between the two sectors, against 4.0% in the baseline and 2.4% of workers holding jobs in both sectors in the data. The common factor is approximated with a five-node Gauss–Hermite quadrature.

Optimal progressivity remains well above the benchmark after adding the estimated transitory component, although by less than in the baseline. This calibration’s benchmark progressivity is 0.018, and the welfare-maximizing level is $\tau^P = 0.060$, about three times the benchmark, under *all three* budget closures, the same closure robustness the baseline optimum displays. The gain at the optimum is 1.85% of consumption under the benchmark closure (1.26% under fixed absolute spending and 0.58% with the transfer fixed, where the optimum is interior and leaves G/Y above its benchmark level), and 1.12% when leisure is excluded. As in the baseline, the optimum coincides with the peak of tax revenue as a share of GDP. Unlike in the baseline, informality does not fall: the informal share of hours rises with progressivity throughout the explored range and is 0.3 percentage points above its benchmark level at the optimum. The reason lies in the benchmark allocation described above: almost every worker with a formal job already works only formal hours, so progressivity has no informal hours to draw into formal work, the margin behind the fall in informality in the baseline. The small increase reflects formal workers reducing their hours by about 1% and a slightly lower job-finding rate, which responds about a third as much as in the baseline because the surplus on a formal match is large. The gain therefore comes from insurance and redistribution, with consumption inequality falling by 1.6%, rather than from formalization. The lower optimum is in line with the higher ξ this calibration selects: as Appendix D.5 shows, informal returns closer to linear in ability make the evasion margin bind at lower progressivity.

C.2 Alternative calibration (b): informal returns fixed at the panel estimate

Alternative calibration (b) replaces both parameters of the informal technology with their panel counterparts (Appendix A.3). The curvature ξ is fixed at the instrumental-variables estimate of the joint slope–intercept specification on lag-matched observations of salaried workers, 0.813 (s.e. 0.037), and is no longer calibrated; since survey earnings are net of the labor tax, the corresponding curvature with respect to gross formal earnings is $0.813(1 - \tau^P) = 0.80$. In addition, the persistence of the efficiency process is set to the panel’s annual estimate, $\rho = 0.944$, instead of the baseline’s 0.98, and σ_{LR} is re-calibrated. This calibration bundles two panel-based changes, ξ , and ρ , rather than a single-parameter deviation; Appendix D.5 reports the effect of moving ξ alone, without re-calibration.

With ξ fixed below one and σ_{LR} free, the calibration keeps much of the baseline inequality fit: σ_{LR} comes back at 1.02, wealth inequality is close to its target (Gini_w 0.70 against 0.72), consumption inequality is somewhat lower (0.38 against 0.41), K/Y is close (2.38 against 2.44), and the model’s formal log-earnings variance (0.328) is close to the panel’s total (0.325, of which 0.193 is persistent). The fit of sectoral earnings and of informality deteriorates. With ξ well below one and a single persistent process common to both sectors, informal earnings compress ability differences rather than amplifying them, so the model can generate an informal variance excess only through sorting and the intensive margin: the persistent variance gap falls to 0.11 (baseline and data 0.17), the mean log-earnings gap overshoots at 0.98 (baseline 0.71, data 0.72), and the targeted wage ratio itself cannot be reached (0.85 against 0.92). The wage ratio trades off against informality: low-productivity workers leave formal work, so the formal pool selects on high productivity, where the informal-to-formal ratio is lower when $\xi < 1$, and the formal wage premium set by the wage rule widens. The calibration reaches a ratio of 0.85 only at an informal share of hours of 0.75, against 0.51 in the data and 0.58 in the baseline; with σ_{LR} near one, a ratio of 0.92 requires an informal share of hours of 0.9 or more. The panel explains why the variance gap is hard to reach. In the panel, the excess persistent dispersion of informal earnings is concentrated among the self-employed, whose earnings pool labor and business income. The informal variance excess in the data is thus a composition effect between two groups, not a property of how a single informal technology maps ability into earnings, and a model with one informal technology cannot carry it once ξ is fixed at the panel estimate.

This calibration also favors substantially higher progressivity. Its benchmark progressivity is 0.016, and the welfare-maximizing level under the benchmark closure is $\tau^P = 0.125$, about eight

times the benchmark, with a CEV of 1.24% (0.090 and 0.43% under the fixed-absolute-spending closure, and 0.110 and 1.19% with the transfer fixed and G/Y at least at its benchmark level). That the optimum lies above the baseline's 0.100 is consistent with the sensitivity analysis of Appendix D.5, where a lower ξ raises the optimum. Informality falls at the optimum, by 7.2 percentage points of the informal share of hours, consumption inequality falls by 2.8%, and measured TFP rises by 0.28%, so the qualitative content of Section 5.1 is unchanged. Two features differ from the baseline. The welfare profile is flat at the top: the CEV varies by less than 0.03 percentage points between $\tau^P = 0.110$ and 0.135, so the location of the optimum within that range is not sharply determined. And tax revenue as a share of GDP peaks at a lower progressivity, $\tau^P \approx 0.06$, and is 0.15 percentage points below its benchmark level at the optimum. The gains are also smaller than in the baseline (4.34%).

D Additional Model Results

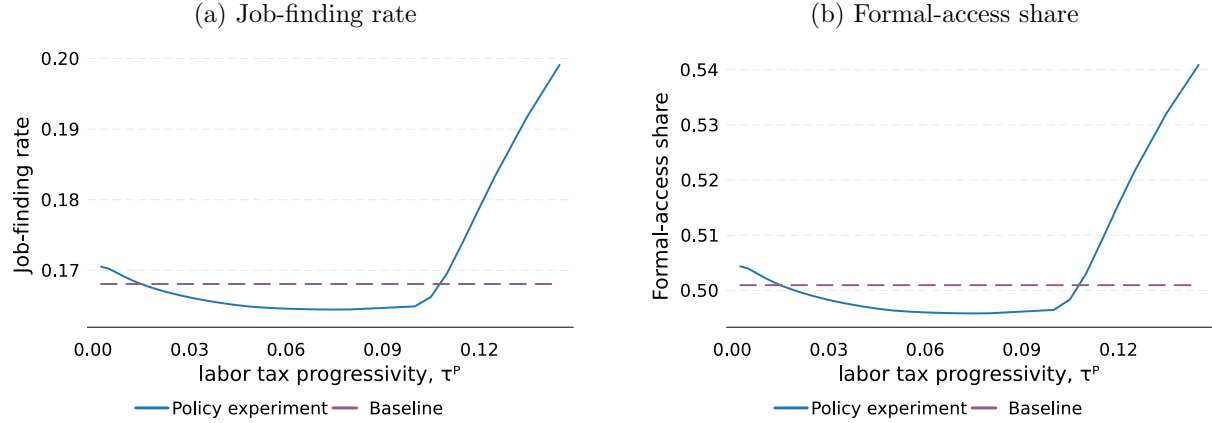
This appendix reports additional results from the quantitative model. Section D.1 reports how job finding and access to formal jobs respond to progressivity. Section D.2 examines the robustness of the optimal-progressivity results to the government budget closure, repeating the main experiment under two alternatives to the benchmark (constant G/Y , transfer-rebated) closure. Section D.4 states the welfare criterion and the consumption-equivalent metric used throughout. Section D.5 reports the sensitivity of the optimum to the wage-rule, labor-supply, and informal-returns parameters. Section D.6 presents results from an economy with homogeneous labor-market transitions, showing that the negative informality profile over the income distribution persists even without skill-specific transition rates. Section D.7 reports the informal share of hours worked by labor-income decile and compares the model's average tax rates, including the tax on business profits, with official incidence estimates. Section D.8 reports comparative statics from the analytical model.

D.1 Labor-Market Frictions and Access to Formal Jobs

Progressivity also changes access to formal jobs, through the vacancy posting of the labor agency. Figure D.1 reports the average job-finding rate into the formal sector and the share of households holding a formal job along the progressivity experiment of Section 5.1. Between the baseline and the optimum, $\tau^P = 0.10$, both are nearly flat: the job-finding rate moves from 0.168 to 0.165, and the formal-access share from 0.501 to 0.496. Up to the optimum, the most productive households keep all their hours in formal work, so formal labor and the labor agency's surplus change little, and the fall in informality comes from the inclusion margin among households that already hold formal jobs.

Beyond the optimum, the evasion margin changes the agency's incentives. As high-productivity households move hours into informal work, effective formal labor becomes scarcer, its price rises, and the agency's surplus per efficiency unit of formal labor, $\Omega p^f - w^f$, increases by 63% between the optimum and the top of the grid, $\tau^P = 0.145$. Vacancy posting responds: the job-finding rate rises by about 21%, and the formal-access share by 4.4 percentage points. Job finding rises for every worker without formal access, so the new formal jobs are spread roughly in proportion to them: households in the bottom 75% of the productivity distribution, 79% of those without access, take 77% of the new jobs. Each newly attached worker supplies most of its hours formally, nearly all of them in the bottom group and about 80% among top earners, so one newly attached top earner

offsets the reallocation of about four top earners already in formal jobs. Over this range, wider access lowers the informal share of hours by 4.3 percentage points, more than offsetting the 2.9 points added by the top earners' reallocation, so the informal share keeps falling, by 1.1 points in total, even as evasion erodes the formal tax base and revenue falls.



Notes: Labor-tax progressivity τ^P varies as in Figure 7: government consumption remains 15% of output, the lump-sum transfer balances the budget, and other tax parameters are fixed. Panel (a) reports the average rate at which households without a formal job find one, $f(\theta)$. Panel (b) reports the share of households holding a formal job, entrepreneurs included. Dashed lines mark the baseline.

Figure D.1: Job finding and access to formal jobs as progressivity varies

D.2 Alternative Government Budget Closures

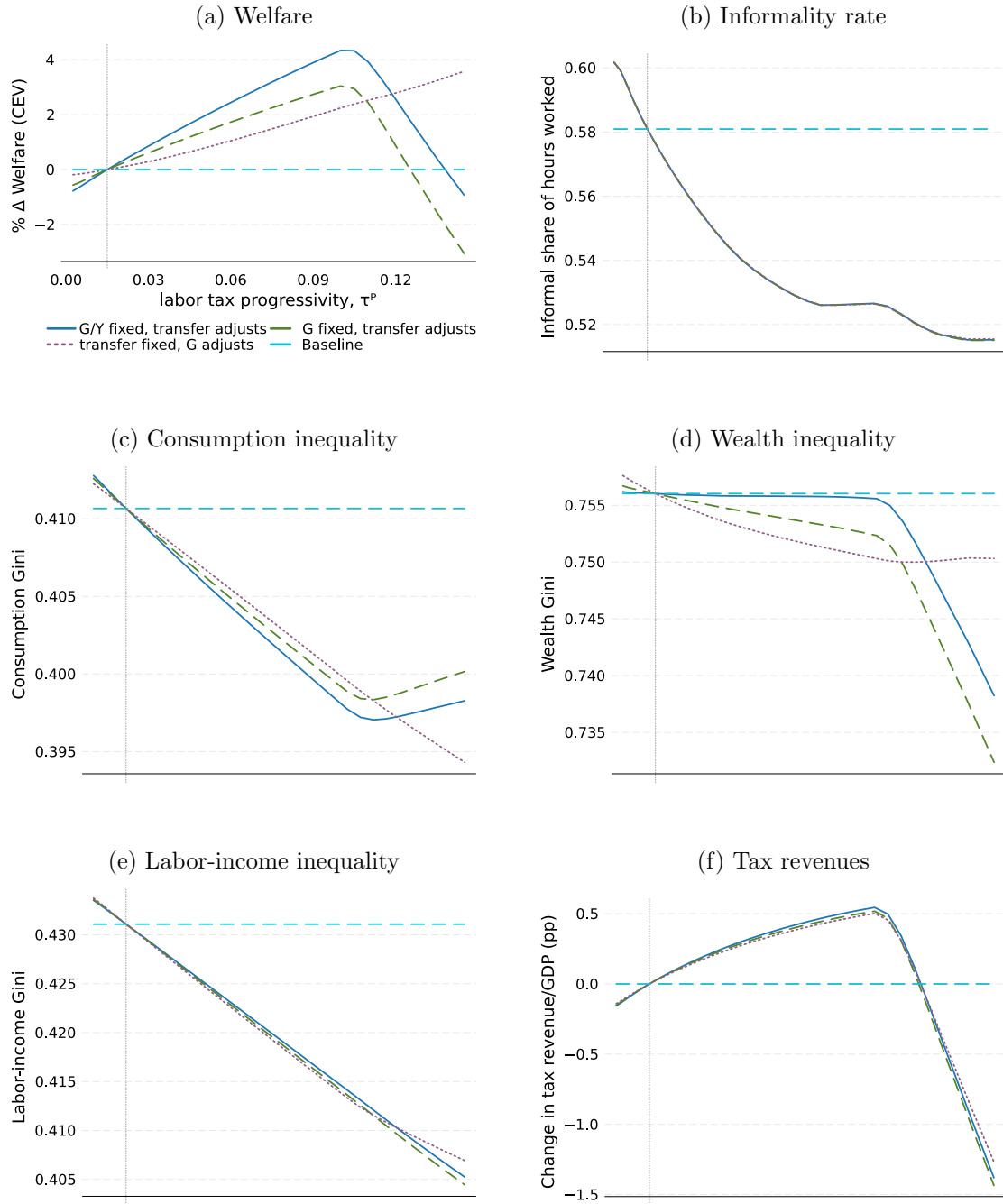
The benchmark experiment holds the spending-to-output ratio at its calibrated $G/Y = 15\%$ and rebates revenue changes to households through the lump-sum transfer. This appendix repeats the progressivity experiment under two alternative closures. In the first, government consumption is held fixed in absolute terms while the transfer continues to adjust; spending then no longer scales with output, so reforms that expand (contract) the economy are not partially absorbed (cushioned) by spending. In the second, the transfer is held fixed at its baseline level and government consumption absorbs revenue changes one for one, so that households receive none of the fiscal dividend of a reform. Figure D.2 reports the resulting profiles alongside the benchmark ones, and Section D.3 extends the comparison to the other fiscal instruments.

The location of the optimum is essentially unchanged across all three closures: the welfare-maximizing progressivity is $\tau^P = 0.100$ under fixed absolute spending and $\tau^P = 0.115$ under the fixed-transfer closure, against 0.100 in the benchmark, and informality at the optimum falls to within

a fraction of a percentage point of its benchmark-closure value in every case. The informality path itself is practically the same under the three rules (Panel D.2b): sector choice responds to the marginal returns set by the tax schedule, while the rules differ only in how revenue is used, as a lump-sum transfer or as government consumption, and the income effects of that difference on labor supply and sector allocation are small. At the top of the grid, the transfer and government consumption differ across the rules by up to 1.9 and 1.7 percentage points of output, yet aggregate labor supply differs by less than 1% and the informal share of hours by 0.04 percentage points. Under the fixed-transfer closure, government consumption absorbs every change in revenue and has no utility value, so welfare keeps rising along the explored grid as government consumption shrinks. We therefore report this closure’s optimum subject to G/Y not falling below its benchmark value of 15%, a constraint that binds: G/Y falls below 15% at $\tau^P \approx 0.12$. Requiring in addition that government consumption not fall in levels gives $\tau^P = 0.105$ and a gain of 2.39%, while requiring in addition that output not fall below its benchmark level leaves the benchmark schedule as the best one: output falls at every progressivity above the benchmark, and G/Y falls below 15% at every progressivity below it. What the closure changes is the size of the gains: 4.3% of consumption in the benchmark, 3.0% with absolute spending fixed, and 2.7% when marginal revenue is absorbed by government consumption that yields no utility. The ordering is instructive. Under the fixed-transfer closure households receive none of the fiscal dividend of formalization, so gains remain even without additional lump-sum resources; the closure comparison shows how the welfare gain depends on the fiscal adjustment accompanying the change in the schedule. Because the location of the optimum is closure-robust while its value is not, we regard the benchmark closure, the budget rule the calibration itself is computed under, as the appropriate basis for the optimal-policy statements in the main text. The transitory-risk calibration of Appendix C passes the same test: its optimum sits at $\tau^P = 0.060$ under all three closures, with gains of 1.85%, 1.26%, and 0.58%, and with the transfer fixed its optimum is interior and leaves G/Y above its benchmark level.

D.3 Fiscal Instruments Under Alternative Closures

Section 5.2 compares progressivity with the two level instruments, the labor-tax level and the capital tax, under the benchmark closure. Table D.1 reports each instrument’s welfare-maximizing reform under all three closures, evaluated at that optimum: the size of the reform, the welfare gain, and the responses of informality, consumption inequality, tax revenue, and the budget item that clears the government budget.



Notes: Progressivity varies under three budget rules: government consumption fixed at 15% of output with the lump-sum transfer adjusting; government consumption fixed in levels with the transfer adjusting; and the transfer fixed with government consumption adjusting. Other tax parameters are fixed. The rules coincide at baseline progressivity $\tau^P = 0.015$ (vertical line); horizontal lines mark baseline outcomes. Welfare gains are steady-state CEVs (percent), including leisure. Panels (b)–(e) show levels. Panel (f) shows changes in tax revenue/GDP in percentage points. Government consumption has no direct utility value. Under the fixed-transfer rule, government consumption falls below 15% of output for τ^P above about 0.12, and welfare keeps rising as it shrinks; the text reports this rule's optimum subject to $G/Y \geq 15\%$, $\tau^P = 0.115$.

Figure D.2: The progressivity experiment under the three government budget closures

Two patterns organize the table. First, for progressivity the closure changes the size of the gains but not the reform: the optimum is +8.5 percentage points of τ^P under both rebate closures and +10.0 when the transfer is fixed, where the G/Y constraint binds; the informal share of hours falls by 5.4–6.0 percentage points in every case, and the welfare gain runs from 4.3% with the rebate and spending indexed to output, to 3.0% with spending fixed, to 2.7% when the additional revenue is absorbed by government consumption. Second, the level instruments live off the rebate. Under the two closures that return revenue to households, raising the labor-tax level or the capital tax finances a demogrant and raises welfare (by up to 6.3% and 1.6% under the benchmark closure), the former at the cost of a 13–20 percentage point rise in the informal share of hours, the latter with the labor-market allocation essentially unchanged and its gains still rising at the edge of the explored range. These gains rest on much larger revenue increases than progressivity’s: at the revenue change of the progressivity optimum, the labor-tax level and the capital tax yield 1.4% and 1.2% under the benchmark closure (1.2% and 0.8% with G fixed), against 4.3% (3.0%) for progressivity. Once the transfer is fixed and revenue changes accrue to government consumption, neither level tax can improve welfare by rising, and cutting either one would lower G/Y below its benchmark value: their constrained optimum is no reform, and at the change in revenue/GDP of the progressivity optimum they lower welfare, by 0.11% and 0.18%. Progressivity is the only instrument whose constrained optimum under this closure is a tax increase. Across closures, increased progressivity combines welfare gains with a substantial reduction in informality. Without the G/Y constraint, small cuts in the level taxes improve welfare only because they shrink government consumption.

D.4 Welfare Criterion

Following the quantitative literature on optimal progressivity (Conesa and Krueger, 2006; Heathcote et al., 2017; Holter et al., 2019), we evaluate reforms with a utilitarian social welfare function over the stationary distribution. Let μ_τ denote the stationary distribution over household states under progressivity τ^P , and let c and ℓ denote the associated policy functions. For the consumption-equivalent calculation, omit the additive constants in utility and define

$$\mathcal{W}^c(\tau^P) = \int \frac{c^{1-\zeta}}{1-\zeta} d\mu_\tau, \quad \mathcal{W}^\ell(\tau^P) = \int \Phi \frac{(1-\ell)^{1-\phi}}{1-\phi} d\mu_\tau. \quad (\text{D.1})$$

Total welfare is $\mathcal{W} = \mathcal{W}^c + \mathcal{W}^\ell$. Omitting constants leaves welfare differences unchanged and makes the consumption component homogeneous. The CEV of reform R relative to benchmark B is the

Table D.1: Welfare-maximizing reforms by instrument and budget closure

Closure	Δ instrument at optimum (pp)	Welfare CEV (%)	Δ informal hours share (pp)	Δ Gini consumption ($\times 100$)	Δ revenue / GDP (pp)	Δ budget margin (pp GDP)	CEV at equal revenue (%)
<i>Progressivity, τ^P</i>							
G/Y fixed	+8.5	4.34	-5.43	-1.29	+0.54	+0.58	4.34
G fixed	+8.5	3.04	-5.44	-1.18	+0.52	+0.22	3.04
Transfer fixed	+10.0 [‡]	2.65	-6.03	-1.30	+0.11	-0.27	2.65
<i>Labor-tax level, $1 - \tau^L$</i>							
G/Y fixed	+10.0	6.33	+19.65	-1.39	+3.28	+3.24	1.42
G fixed	+7.0	4.78	+13.29	-0.96	+2.64	+2.32	1.17
Transfer fixed	—	—	—	—	—	—	-0.11
<i>Capital tax, τ^k</i>							
G/Y fixed	+9.0 [†]	1.62	-0.05	-0.24	+0.77	+0.78	1.16
G fixed	+9.0 [†]	1.09	-0.04	-0.19	+0.75	+0.63	0.76
Transfer fixed	—	—	—	—	—	—	-0.18

Notes: Each row maximizes welfare over the explored range for one instrument under the stated budget rule; other tax parameters are fixed. The first column names what the rule holds fixed: with G/Y or G fixed, the lump-sum transfer adjusts; with the transfer fixed, government consumption adjusts. Changes are relative to the baseline. CEV denotes steady-state consumption-equivalent variation, including leisure. The labor-tax level $1 - \tau^L$ is the average tax rate at mean formal earnings. The Gini column reports 100 times the absolute change. The budget margin is the change in the transfer or government consumption, divided by baseline output and multiplied by 100. Government consumption has no direct utility value. With the transfer fixed, welfare is maximized subject to government consumption remaining at least at its baseline share of output. The last column reports the instrument's CEV at the reform, nearest to the baseline, that changes revenue/GDP by as much as the progressivity optimum under the same closure (+0.54 percentage points with G/Y fixed, +0.52 with G fixed, +0.11 with the transfer fixed), interpolated linearly on the instrument's grid; with the transfer fixed, this reform is not the two level instruments' constrained optimum, which is no reform. [†] marks a grid boundary, not an interior optimum; [‡] marks an optimum at which the requirement that government consumption stay at least at 15% of output binds. — marks no reform: the best reform is the baseline itself. Explored ranges: $\tau^P \in [0.0025, 0.145]$, $\tau^L \in [0.755, 0.885]$, and $\tau^k \in [0.071, 0.171]$.

uniform proportional change Δ in benchmark consumption satisfying

$$(1 + \Delta)^{1-\zeta} \mathcal{W}_B^c + \mathcal{W}_B^\ell = \mathcal{W}_R, \quad \Delta = \left(\frac{\mathcal{W}_R - \mathcal{W}_B^\ell}{\mathcal{W}_B^c} \right)^{1/(1-\zeta)} - 1. \quad (\text{D.2})$$

With $\zeta = 2$, the exponent is -1 . Quantitative-model welfare figures and tables report Δ in percent, relative to the benchmark of the same fiscal closure and, in the inequality experiment, the same σ_{LR} -economy. These comparisons account for consumption and leisure in the stationary distributions; they do not include the transition between steady states.

At the welfare-maximizing progressivity of the benchmark experiment the two components move in the same direction. The optimal reform shifts hours out of informal work, about four fifths of them into formal work and the rest into leisure, as aggregate hours fall slightly, so the leisure component contributes about 1.6 percentage points of the 4.3% gain, and its contribution is flat in the neighborhood of the optimum, which is set by the consumption component.

D.5 Sensitivity: Wage Rule, Labor Supply Elasticity, and Informal Returns

Table D.2 reports the welfare-maximizing progressivity when one structural parameter at a time is moved away from its baseline value, holding all remaining parameters fixed. The optimum is insensitive to the wage-rule weight η , which governs how the surplus between the intermediate-goods price and the informal wage is split: moving η by ± 0.15 around its baseline value of 0.60 shifts $\tau^{P,*}$ by 0.005 to 0.01, so the results do not hinge on how the surplus is divided. The Frisch elasticity matters through the strength of the labor-supply response itself: doubling it to 0.60 lowers the optimum to 0.09 with larger gains, while halving it to 0.15 raises the optimum to 0.12 and halves the gain: with less responsive labor supply, progressivity has less formalization to gain, so the stakes fall, and less evasion to fear, so the planner can push the schedule further.

The curvature of informal returns to skill, ξ , moves the location of the optimum the most: lowering it by 0.05 raises the optimum to 0.135 at an unchanged gain (informal earnings compress faster at the top, so evasion is more expensive and the planner can push progressivity further), while raising it to 0.968 (informal returns closer to linear in skill) pulls the optimum down to 0.07 with about 70% of the gain, as the evasion margin bites earlier. The Frisch elasticity moves the size of the gain more than ξ does, between 2.11% and 4.99% against 3.11% and 4.35%. The sensitivity of the optimum to ξ is why ξ is disciplined directly in the calibration through the sectoral earnings moments, and why Appendix C reports two alternative calibrations that discipline ξ differently: the

transitory-risk calibration, where it settles at 0.98, and the calibration with ξ fixed at the ENOE panel estimate.

Table D.2: Sensitivity of the welfare-maximizing progressivity $\tau^{P,*}$

Parameter	Value	G/Y fixed		G fixed	
		$\tau^{P,*}$	CEV at $\tau^{P,*}$ (%)	$\tau^{P,*}$	CEV at $\tau^{P,*}$ (%)
– (baseline)	–	0.100	4.34	0.100	3.04
Wage rule η	0.45	0.095	4.11	0.095	2.86
	0.75	0.110	4.63	0.110	3.23
Frisch elasticity	0.15	0.120	2.11	0.120	1.49
	0.60	0.090	4.99	0.090	3.06
Informal curvature ξ	0.868	0.135	4.35	0.125	2.72
	0.968	0.070	3.11	0.070	2.18

Notes: One parameter varies at a time around the baseline values: wage-rule surplus share $\eta = 0.60$, Frisch elasticity 0.30, and informal-return curvature $\xi = 0.918$. Remaining parameters are not recalibrated; the Frisch elasticity is varied through the curvature ϕ of the utility of leisure, holding its weight Φ fixed. CEV denotes steady-state consumption-equivalent variation, including leisure, relative to each perturbed economy’s own equilibrium at baseline progressivity $\tau^P = 0.015$. The lump-sum transfer balances the budget, with government consumption fixed at 15% of output or at its baseline level. Optima are selected over the explored progressivity grid, $\tau^P \in [0.0025, 0.145]$.

D.6 Homogeneous Labor-Market Transitions

Figure D.3 reports results for an economy in which all workers face identical formal-sector labor market dynamics. Panels D.3c and D.3d show the flat profiles of separation and job-finding rates across the labor-income distribution, confirming the absence of skill-dependent transition probabilities. As a consequence, the composition of households with and without a formal job changes, as shown in Panel D.3b: households without a formal job become more productive on average, and those with one less productive. Low-skilled households now face more favorable transition rates and the opposite happens to high-skilled ones; this improves labor productivity in the informal sector and lowers it in the formal one. In equilibrium, free entry of vacancies raises the common job-finding rate to 0.191, above the baseline average of 0.168, so the share of households with a formal job rises from 50.1% to 53.3% and the informal share of hours falls from 58.1% to 53.7%.

Even under these homogeneous labor market conditions, the model continues to generate a downward-sloping informality rate over the labor-income distribution, although a flatter one: the informal share of hours falls by about 5 percentage points per decile on average, against 8 in the

baseline and in the data. This outcome reflects the optimal labor supply and sectoral allocation implied by households' intratemporal first-order conditions (17). Sectoral labor supply responds to household-specific incentives, particularly among low-productivity workers, whose higher wealth-effect elasticities amplify the reallocation margin between formal and informal activities; households at the lowest productivity level work informally even when they hold a formal job. In contrast, the profile of informality flattens in the upper half of the income distribution, where the absence of skill-specific transition rates limits further reallocation toward the formal sector: from the fifth to the ninth decile, the informal share of hours falls by 18 percentage points, against 23 in the baseline.

D.7 Additional Fit Checks

Panel 6c in Section 4.7 measures informality as the share of workers in informal jobs. Panel D.4a measures it as the informal share of hours worked, by labor-income decile, in the baseline economy and in the data.

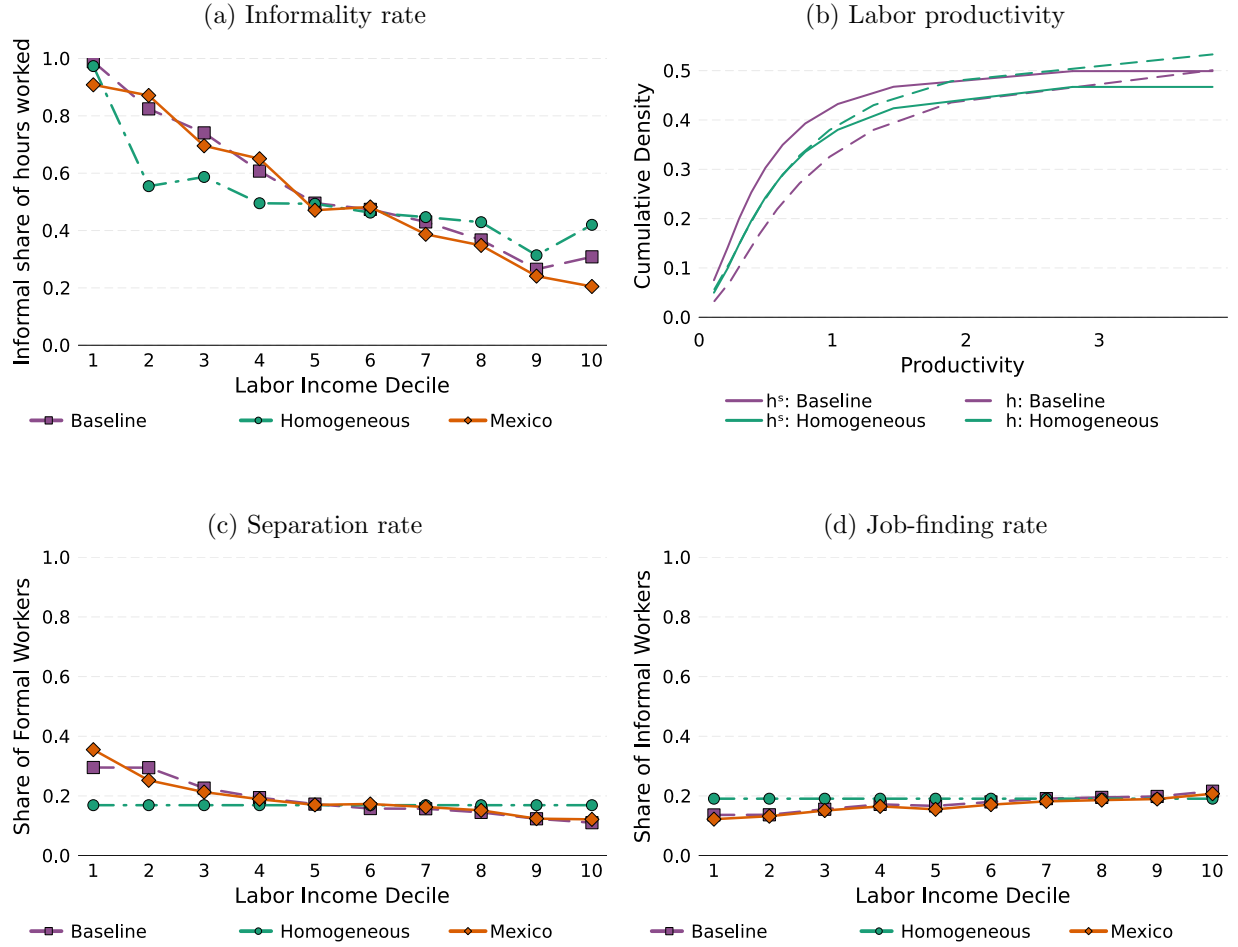
Panel 6d in Section 4.7 compares the model's average labor-tax rate by decile of household income, excluding entrepreneurs, with the incidence estimates of Mexico's Ministry of Finance (SHCP) for the income tax on salaries. Panel D.4b brings the entrepreneurs back. The model series adds the tax on the profits that entrepreneurs receive, $\tau^{\text{profit}} \Pi(h)$, to each household's labor tax and divides by total gross household income; the data series is the personal income tax paid by individuals, which also covers their business income, as a share of total gross household income, from the same SHCP report. Because only entrepreneurs receive profits, adding the profit tax changes the model's profile only in the deciles where the entrepreneurs are. The model tracks the data about as closely as in Panel 6d: the root-mean-squared gap across the ten deciles is 1.2 percentage points, against 1.1 there.

D.8 Comparative Statics in the Analytical Model

This section reports comparative statics from the analytical model of Section 3, examining how structural features of the economy shape optimal progressivity.

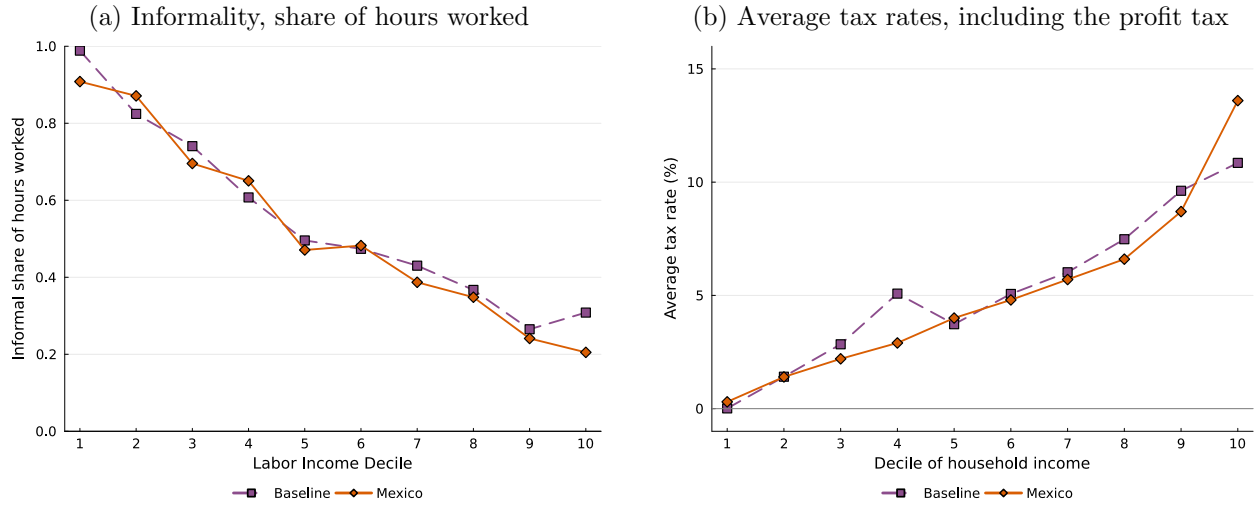
Higher public financing needs and higher unemployment risk raise informality, reduce welfare, and lower optimal progressivity. With lower spending or unemployment risk, optimal progressivity rises to 0.28–0.31.

Transfers provide income to low-ability households and insurance against formal-job loss. Rais-



Notes: In the homogeneous economy every formal job separates at rate 0.169, the baseline average, and every household without a formal job finds one at rate 0.191, the equilibrium job-finding rate of this economy; no parameter is recalibrated. Panel (a) reports the informal share of hours worked, and panels (c) and (d) the transition rates, by labor-income decile. Panel (b) reports cumulative distributions of productivity, h^s for households without a formal job and h for households with one, each scaled to its group's population share.

Figure D.3: Informality rate, productivity distribution, and formal-sector transition rates



Notes: Panel (a) reports the informal share of hours worked, by decile of the worker's own labor income, in the baseline economy and in Mexico's National Survey of Occupation and Employment (ENOE, 2024). Panel (b) reports average tax rates, in percent, by decile of household income. Baseline ($\tau^P = 0.015$): labor tax plus the tax on business profits, $\tau^{\text{profit}} \Pi(h)$, divided by total gross household income (labor, capital, and profit income); the tax on capital income, $\tau^k ra$, is not included. Mexico: personal income tax paid by individuals (*ISR personas físicas*) divided by total gross household income, by decile of households, as estimated by the SHCP from ENIGH 2024 in the scenario that assigns the tax only to earners affiliated with social security (same report as Panel 6d).

Figure D.4: Informal share of hours and average tax rates including the profit tax, by decile

Table D.3: Effects of progressive fiscal policy: comparative statics

	Optimal policy		Welfare	Ex-post Gini	Informality		
	τ^P	τ^L			Average	Bottom 25%	Top 25%
Baseline	0.24	0.834	-0.839	0.468	0.542	0.837	0.259
A: Government expenditure to GDP ratio, G/Y (Baseline 0.15)							
0.10	0.28	0.893	-0.782	0.431	0.494	0.746	0.269
0.20	0.20	0.762	-0.899	0.501	0.606	0.920	0.259
B: Unemployment risk, q (Baseline 0.06)							
0.01	0.31	0.879	-0.701	0.396	0.400	0.705	0.111
0.10	0.21	0.803	-0.914	0.504	0.625	0.898	0.344
C: Lump-sum transfers as a fraction of GDP, τ^T/Y (Baseline 0.0)							
0.01	0.23	0.821	-0.830	0.467	0.550	0.857	0.250
0.02	0.22	0.808	-0.823	0.463	0.560	0.876	0.242

Notes: Each row reports outcomes at that economy's welfare-maximizing progressivity τ^P . The tax scale τ^L balances the budget, financing the specified government consumption and transfers. Welfare is population-average expected utility, including labor disutility, in levels. The ex-post Gini measures labor-earnings inequality. Gini coefficients and informality rates are fractions. Other structural parameters remain at the static-model baseline.

ing transfers to 1% or 2% of GDP increases welfare, lowers optimal progressivity slightly, and makes earnings more equal.

E Details and Additional Results of the Static Model

E.1 Equilibrium Conditions

The government levies taxes on formal earnings to finance spending G and lump-sum transfers τ^T . Its balanced budget is

$$G + \tau^T = (1 - q) \int_0^\infty \left(w^f h n \varpi - \tau^L (w^f h n \varpi)^{1-\tau^P} \right) f_h(h) dh, \quad (\text{E.1})$$

where f_h is the density of ability. In the baseline experiments, $\tau^T = 0$ and τ^L adjusts to keep $G/Y = 0.15$.

Definition 1 *An equilibrium consists of allocations $\{n, \varpi, c^f, c^s, L^s, L^f, Y\}$ and prices $\{w^f, w^s\}$ such that:*

- *Households choose n and ϖ to maximize utility given wages, the tax schedule (τ^L, τ^P) , transfers τ^T , and unemployment risk q .*
- *Firms maximize profits given wages.*
- *The government keeps a balanced budget.*
- *Markets clear:*

$$Y = (1 - q) \int_0^\infty c^f f_h(h) dh + q \int_0^\infty c^s f_h(h) dh + G, \quad (\text{E.2})$$

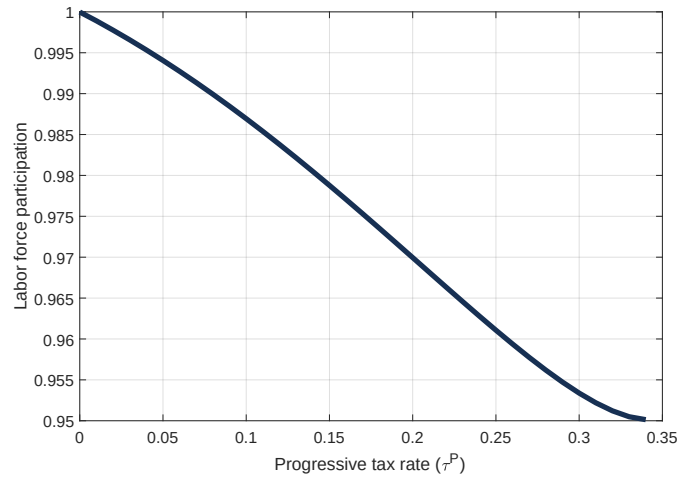
$$L^f = (1 - q) \int_0^\infty h n \varpi f_h(h) dh, \quad (\text{E.3})$$

$$L^s = \int_0^\infty \Xi h^\xi n (1 - \varpi) f_h(h) dh. \quad (\text{E.4})$$

The net marginal return to reallocating labor toward formality, conditional on finding a job, is

$$\tilde{w} = (1 - \tau^P) \tau^L (w^f h n)^{1-\tau^P} \varpi^{-\tau^P} - w^s \Xi h^\xi n. \quad (\text{E.5})$$

At an interior allocation, equation (7) equates $(1 - q)\tilde{w}/c^f$ to the expected marginal utility loss $q w^s \Xi h^\xi n / c^s$ when no formal job is found.



Notes: Static-model equilibria as tax progressivity varies. Lump-sum transfers are zero, and the tax scale adjusts to finance government spending of 15% of output. Other structural parameters are held fixed.

Figure E.1: Progressivity and total labor supply

E.2 Additional Fiscal-Policy Results

Figure E.1 complements the welfare and informality profiles in Figure 5. Total labor supply responds relatively little to progressivity: increasing τ^P from zero to 0.35 reduces labor supply by about 5%. Most adjustment therefore occurs through the allocation of work across sectors.

F Additional Empirical Evidence

F.1 Survey Coverage and Earnings Inequality

The four surveys are carried out continuously over the year. The number of households interviewed was 708,340 in Brazil, 277,158 in Colombia, 496,153 in Mexico, and 33,886 in Peru. Each survey is representative at the national, urban, and rural levels and is an official source of labor-market statistics. For Mexico, the official source of income inequality statistics is the National Household Income and Expenditure Survey (ENIGH), which provides more detailed information on income sources but less information on labor-market outcomes than the ENOE.

Brazil and Mexico interview households for five consecutive quarters. Peru follows a subsample annually for up to five years, while Colombia includes retrospective labor-market questions. Appendix A describes the surveys and the definitions used in the cross-country comparisons.

Table F.1 reports inequality in household per-capita labor earnings. Inequality is substantial in all four countries, and the percentile ratios show that the bottom of the distribution lies particularly far below the median. The Gini ranges from 0.424 in Mexico to 0.580 in Colombia.

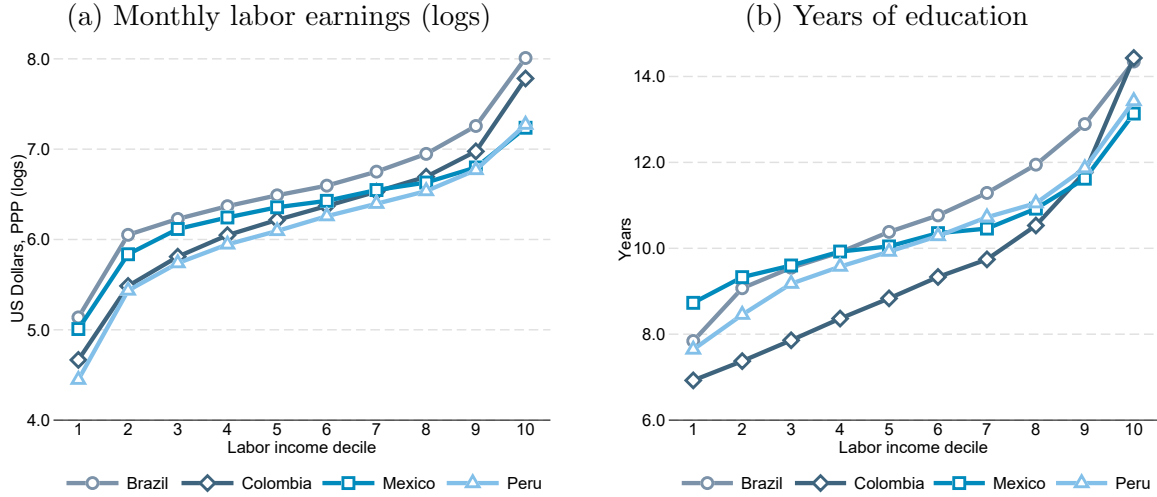
	Brazil	Colombia	Mexico	Peru
Gini	0.543	0.580	0.424	0.498
Variance of logs	1.145	1.303	0.815	1.164
Percentile ratio 90/10	13.200	16.833	7.998	13.339
Percentile ratio 75/25	3.500	4.127	2.907	3.463
Percentile ratio 90/50	3.451	4.040	2.544	3.159
Percentile ratio 10/50	0.261	0.240	0.318	0.237
Survey year	2024	2023	2024	2023

Notes: Household per-capita labor earnings equal total household labor earnings divided by household size. The sample includes households with positive labor earnings. Earnings cover the main job in Colombia and Mexico and all jobs in Brazil and Peru; Peru’s annual earnings are divided by 12. Estimates use survey weights. Appendix A describes the surveys.

Table F.1: Inequality measures, households’ per capita labor earnings

F.2 Earnings and Education Across the Income Distribution

Figure F.1 shows monthly labor earnings in logs and average years of education by labor-income decile. Earnings are measured in PPP-adjusted US dollars. Earnings differences are largest at the



Notes: The sample covers individuals aged 15 and older in households with positive labor earnings, grouped into country-specific deciles of household per-capita labor earnings. Panel (a) shows employed individuals' monthly earnings in logs, measured in PPP-adjusted US dollars for the survey year; panel (b) shows average completed years of schooling. Earnings cover the main job in Colombia and Mexico and all jobs in Brazil and Peru. Survey years are 2024 for Brazil and Mexico and 2023 for Colombia and Peru. Estimates use survey weights; Appendix A describes the surveys.

Figure F.1: Monthly earnings (logs) and average years of education by labor income decile, 2023/24

ends of the distribution. Consistent with its lower earnings inequality in Table F.1, Mexico displays a flatter gradient and lower earnings in the top decile than the other countries.

Schooling rises with income in all four countries. Colombia has lower average schooling (9.1 years), but schooling in its top decile is comparable with Brazil and Peru. As with earnings, the schooling gradient in Mexico is flatter, including at the top.

F.3 Additional Labor-Market Comparisons

The unemployment gradients in Figure 1 differ across countries, with larger differences between the bottom and top in Brazil and Colombia. These comparisons use individuals aged 15 and older in households with positive labor earnings, ranked by household per-capita labor earnings. The employment rate is the employed-to-population ratio, the unemployment rate is the unemployed-to-labor-force ratio, and the informality rate is the informal-to-employed ratio.

G Details of the Quantitative Model

This appendix supplies the factor-price equations, equilibrium definition, and income-process annu-
alization for Section 4.

G.1 Factor Prices and Aggregation

Under perfect competition, net of depreciation δ , rental rate of capital r and the price of intermediate
inputs p^H are given by:

$$r + \delta = \alpha \nu \frac{Y}{K}, \quad p^H = (1 - \alpha) \nu \frac{Y}{H}. \quad (\text{G.1})$$

From (10), the relative price of formal intermediate inputs p^f and the return per efficiency unit
of informal labor w^s satisfy:

$$p^f = (1 - \alpha) \nu \frac{Y}{H} \left(\frac{H}{\Omega L^f} \right)^{1/\varepsilon}, \quad (\text{G.2})$$

$$w^s = \varkappa p^f \left(\frac{\Omega}{\varkappa} \cdot \frac{L^f}{L^s} \right)^{1/\varepsilon}. \quad (\text{G.3})$$

Aggregation. Combining (9) and (10), we can express the aggregate production function as:

$$Y = \left\{ A \left[(\Omega(1 - \lambda^s))^{\frac{\varepsilon-1}{\varepsilon}} + (\varkappa \lambda^s)^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon \nu (1-\alpha)}{\varepsilon-1}} \right\} (K^\alpha L^{1-\alpha})^\nu, \quad (\text{G.4})$$

where $L = (L^f + L^s)$ and measured TFP depends on the informal share in efficiency units $\lambda^s \equiv \frac{L^s}{L}$.

Equation (G.4) makes the model's TFP-informality link explicit: measured TFP is

$$\text{TFP}(\lambda^s) = A \left[(\Omega(1 - \lambda^s))^{\frac{\varepsilon-1}{\varepsilon}} + (\varkappa \lambda^s)^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon \nu (1-\alpha)}{\varepsilon-1}}. \quad (\text{G.5})$$

For $\varepsilon > 1$, this expression decreases with λ^s when $\lambda^s > \varkappa^{\varepsilon-1} / (\Omega^{\varepsilon-1} + \varkappa^{\varepsilon-1})$. Reallocation changes
measured TFP through the composition of production; Section 5 reports its magnitude across the
policy experiments.

G.2 Stationary Equilibrium

A stationary equilibrium consists of household value and policy functions $\{V, g^a, g^\ell, g^\kappa, g^c\}$ over
states (a, h, F) , aggregate inputs $\{K, L^s, L^f\}$, and prices $\{r, p^H, p^f, w^s, w^f\}$. It also includes the

labor agency's value $J(L^f)$ and vacancies V , the job-finding rate f , labor-market tightness θ , fiscal instruments $\{\tau^c, \tau^k, \tau^{\text{profit}}, \tau^L, \tau^P, \tau^T\}$, and an invariant household distribution $\mu(a, h, F)$, such that:

1. households solve their optimization problem given prices, labor market transition rates, and fiscal tools,
2. factor prices satisfy (G.1), (G.3), (G.2), and (20),
3. the zero-profit condition for the labor agency (19) holds,
4. the government budget satisfies (21),
5. the distribution $\mu(a, h, F)$ is stationary and satisfies all agents' policy functions and aggregate consistency conditions,
6. all markets clear.

Market clearing. The formal-hours share is $g^\kappa(a, h, 0) = 0$ for households without formal job access. Market clearing requires:

$$\begin{aligned}
K &= \int g^a(a, h, F) d\mu(a, h, F), \\
L^f &= \int g^\kappa(a, h, F) g^\ell(a, h, F) \cdot h^f d\mu(a, h, F), \\
L^s &= \int [1 - g^\kappa(a, h, F)] g^\ell(a, h, F) \cdot h^s d\mu(a, h, F), \\
Y &= \int \frac{\Pi(h)}{1 - \nu} d\mu(a, h, F)
\end{aligned} \tag{G.6}$$

G.3 Annualization of the Income Process

Income-process parameters are annualized from quarterly values $\rho_q = 0.995$ and $\sigma_q^2 = 0.01$. The corresponding annual persistence is $\rho = \rho_q^4 = 0.980$, and the annual innovation standard deviation is $\sigma_h = \sigma_q \sqrt{1 + \rho_q^2 + \rho_q^4 + \rho_q^6} = 0.199$. These values imply long-run dispersion $\sigma_{LR} = \sigma_h / \sqrt{1 - \rho^2} = 1.00$.